Supplementary Material for "Latent Truth Without a Gold Standard: Multi-Reference Credibility Modeling and Claim Governance for Gene-Regulatory-Network Benchmarking"

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Every table and figure in this supplement is generated by a script of the code repository from its stored analysis tables. Rows, columns or panels marked with a dagger (†) come from analyses defined after the held-out data had been examined (post hoc). Section, table, figure and equation numbers without an "S" refer to the main text.

S1 Data provenance

The evidence annotators of Section 4.2, with the channel through which their records were obtained, their versions, citations and resource URLs (Table S1), and their coverage of the four universes under the curated-conditional observation model (Table S2).

Table S1: Evidence annotators: primary resource, modality family, channel through which the records were obtained, version, citation, resource URL and size (edges, TFs and targets after normalisation to approved HGNC symbols and Lambert TFs).

Resource	Channel	Version	Citation	Resource URL	Edges	TFs	Targets
Literature							
ExTRI	C	as integrated in CollecTRI v1 (2023)	Vazquez et al. (2022)	`http://www.extri.org` (offline)	25,919	841	4,745
NTNU.Curated	C	as integrated in CollecTRI v1 (2023)	Müller-Dott et al. (2023)	`https://github.com/saezlab/CollecTRI`	8,992	571	2,583
GEREDB	C	as integrated in CollecTRI v1 (2023)	Huang et al. (2019)	`http://www.thua45.cn/geredb` (offline)	7,382	593	2,409
TRRUST	C+D+T	TRRUST v2 (2018)	Han et al. (2018)	`https://www.grnpedia.org/trrust/`	7,051	522	2,324
TRED	D	TRED 2007 via RegNetwork 2015, via DoRothEA	Jiang et al. (2007)	`http://rulai.cshl.edu/TRED` (offline)	6,626	127	2,873
TFactS	C+D	as integrated in DoRothEA (2019) and CollecTRI v1 (2023)	Essaghir et al. (2010)	`http://www.tfacts.org` (offline)	2,892	194	1,547
HTRIdb	C+D	v052016, via DoRothEA	Bovolenta et al. (2012)	`http://www.lbbc.ibb.unesp.br/htri` (offline)	2,231	267	1,400
SIGNOR	C	as integrated in CollecTRI v1 (2023)	Perfetto et al. (2016)	`https://signor.uniroma2.it/`	2,090	457	1,093
Pavlidis2021	C	as integrated in CollecTRI v1 (2023)	Chu et al. (2021)	`https://github.com/PavlidisLab/dtri-curation`	1,298	188	770
PAZAR	D	as integrated in DoRothEA (2019)	Portales-Casamar et al. (2009)	`http://www.pazar.info` (offline)	1,289	52	1,093
KEGG	D	as integrated in DoRothEA (2019)	Kanehisa et al. (2012)	`https://www.kegg.jp/`	1,223	142	581
CytReg	C	as integrated in CollecTRI v1 (2023)	Carrasco Pro et al. (2018)	`https://cytreg.bu.edu`	1,202	202	106
IntAct	C+D	as integrated in DoRothEA (2019) and CollecTRI v1 (2023)	Orchard et al. (2014)	`https://www.ebi.ac.uk/intact/`	729	143	423
GOA	C	as integrated in CollecTRI v1 (2023)	Huntley et al. (2015)	`https://www.ebi.ac.uk/GOA/`	605	304	431
ORegAnno	D	ORegAnno 3.0, via DoRothEA	Lesurf et al. (2016)	`https://oreganno.org`	464	65	396
TFe	D	as integrated in DoRothEA (2019)	Yusuf et al. (2012)	`http://www.cisreg.ca/tfe` (offline)	464	45	382
FANTOM4	D	FANTOM4, via DoRothEA	FANTOM Consortium and Riken Omics Science Center (2009); Kawaji et al. (2011)	`https://fantom.gsc.riken.jp/4/`	370	137	105
Other literature	D	DoRothEA-reviews: DoRothEA (2019)	García-Alonso et al. (2019); Gronostajski et al. (2011)	`https://saezlab.github.io/dorothea/`	247	7	240
Motif							
HOCOMOCO	D	HOCOMOCO v11, via DoRothEA (2019)	Kulakovskiy et al. (2018)	`https://hocomoco11.autosome.org/`	9,433	254	5,324
JASPAR	D	JASPAR 2018, via DoRothEA (2019)	Khan et al. (2018)	`https://jaspar.elixir.no/`	8,377	248	4,972

Table S1: Evidence annotators: primary resource, modality family, channel through which the records were obtained, version, citation, resource URL and size (edges, TFs and targets after normalisation to approved HGNC symbols and Lambert TFs). (continued)

Resource	Channel	Version	Citation	Resource URL	Edges	TFs	Targets
Coexpression							
ARACNe-GTEx	D	GTEx v6, via DoRothEA (2019)	Margolin et al. (2006); García-Alonso et al. (2019)	`https://gtexportal.org`	10,931	337	5,903
ChIP (held out)							
PAZAR-HT	D	as integrated in DoRothEA (2019)	Portales-Casamar et al. (2009)	`http://www.pazar.info` (offline)	152,526	24	18,220
ReMap	D	ReMap 2018, via DoRothEA (2019)	Chèneby et al. (2018)	`https://remap.univ-amu.fr/`	117,834	296	11,141
ENCODE-proximal	O	OmniPath TF–target attachment (truncated to DoRothEA/CollecTRI edges)	Gerstein et al. (2012)	`http://encodenets.gersteinlab.org`	6,393	84	4,168
ENCODE-distal	O	OmniPath TF–target attachment (truncated to DoRothEA/CollecTRI edges)	Gerstein et al. (2012)	`http://encodenets.gersteinlab.org`	2,181	69	952

Channel: how the records were obtained, through OmniPath queries of CollecTRI (C, accessed 2026-10-07) or of DoRothEA confidence levels A–D (D, omnipath Python client, accessed 2026-01-08), the TRRUST v2 human file (T), or OmniPath TF–target evidence attached to DoRothEA and CollecTRI records (O). Other literature: the DoRothEA curated reviews and NFIRegulomeDB, which have too few records to model separately. Offline: the resource URL did not respond when the data were accessed (printed without a link); the records come from the integrators. ORegAnno: the published URL `http://www.oreganno.org` no longer resolves (checked 2026-10-15); `https://oreganno.org` leads to the UCSC Genome Browser ORegAnno track. DOIs, family-assignment criteria and processing notes are in the source table of the code repository, and the checksums of the downloaded integrator and TRRUST files in its input manifest.

Table S2: Coverage of the held-in annotators under the curated-conditional observation model in each universe: TFs covered, edges observed (share of the universe) and reports (share of observed edges).

		Global universe					Immune		Kidney		Lung	
Annotator	Family	TFs	Observed	%	Reports	% rep.	obs. %	rep. %	obs. %	rep. %	obs. %	rep. %
ExTRI	LIT	841	4,139,536	44.4	25,919	0.6	33.0	1.3	35.0	1.4	37.5	1.0
NTNU.Curated	LIT	571	1,598,852	17.2	8,992	0.6	17.7	1.1	17.3	1.1	21.0	0.9
GEREDB	LIT	593	1,548,971	16.6	7,382	0.5	18.2	0.8	17.2	0.9	21.3	0.7
TRRUST	LIT	522	1,323,309	14.2	7,051	0.5	16.0	0.9	15.8	0.8	18.0	0.7
SIGNOR	LIT	457	620,651	6.7	2,090	0.3	7.8	0.5	7.0	0.4	8.1	0.5
HTRIdb	LIT	267	419,212	4.5	2,231	0.5	4.1	0.7	4.1	0.7	4.7	0.6
TRED	LIT	127	369,700	4.0	6,626	1.8	3.9	3.1	4.0	2.8	3.2	1.9
TFactS	LIT	194	317,586	3.4	2,892	0.9	4.5	1.3	3.8	1.0	4.1	1.1
GOA	LIT	304	196,423	2.1	605	0.3	2.5	0.5	1.7	0.5	2.5	0.5
Pavlidis2021	LIT	188	162,059	1.7	1,298	0.8	3.3	1.0	2.2	1.7	2.9	1.2
KEGG	LIT	142	95,997	1.0	1,223	1.3	1.2	2.5	1.4	2.6	1.0	1.4
IntAct	LIT	143	76,221	0.8	729	1.0	1.0	1.1	1.5	1.4	1.2	1.0
CytReg	LIT	202	61,210	0.7	1,202	2.0	1.4	3.1	1.8	4.4	1.4	2.9
PAZAR	LIT	52	58,761	0.6	1,289	2.2	0.3	0.8	0.3	10.9	0.2	5.3
ORegAnno	LIT	65	29,187	0.3	464	1.6	0.4	1.4	0.4	0.4	0.2	1.4
FANTOM4	LIT	137	26,992	0.3	370	1.4	0.4	1.0	0.6	1.4	0.3	1.0
TFe	LIT	45	18,811	0.2	464	2.5	0.2	3.0	0.1	16.9	0.3	3.0
Other literature	LIT	7	1,715	0.0	247	14.4	0.0	2.9	0.0	0.0	0.0	–
HOCOMOCO	MOTIF	243	15,200	0.2	862	5.7	0.2	5.3	0.3	5.5	0.2	5.4
JASPAR	MOTIF	232	14,214	0.2	790	5.6	0.2	4.4	0.3	5.8	0.2	5.4
ARACNe-GTEx	COEXP	301	15,952	0.2	1,392	8.7	0.2	15.3	0.3	17.2	0.2	21.8
Any annotator		935	5,084,891	54.6	43,444	0.9	39.6	1.7	41.6	1.8	44.2	1.4

Observed (%): share of universe edges the annotator observes; reported (%): share of its observed edges it reports. Any annotator: edges observed by at least one annotator, and edges with at least one report.

S2 Model fit, identifiability and calibration

Annotator parameters of the global Family-HB fit, label-free model comparison by cross-validation and posterior predictive checks (Section 5.1), and the simulations from the fitted model: recovery and calibration of credibility, and selection of the posterior mode and its consequences for the claim tiers. All simulation results are post hoc.

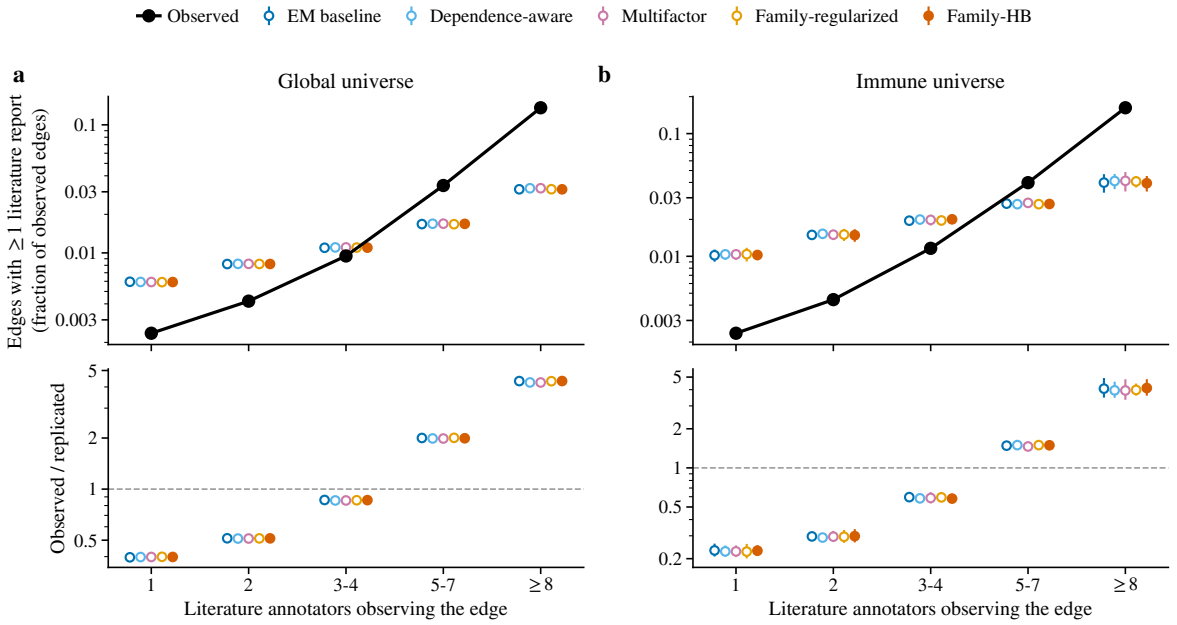


Figure S1: Posterior predictive check of the literature block. For edges observed by a given number of literature annotators (x axis), the fraction with at least one literature report: observed (black) and the mean over 20 data sets replicated from each fitted estimator on the real observation masks (coloured; bars: mean ± 1.96 replicate SD). Lower panels: observed divided by replicated (reference line at 1). Every estimator, including Family-HB, underpredicts the reporting rate of edges observed by many annotators and over-predicts it for edges observed by few (inference: observation is not ignorable).

Table S3: Annotator parameters of Family-HB in the global universe (dominant-mode NUTS posterior means with 95% credible intervals): sensitivity α_j, false-positive rate β_j, burst reporting rate γ_j and the marginal positive likelihood ratio (a report under T against the mixture of all non-T classes), ranked by the latter.

Rank[†]	Annotator	Family	Observed	Reports	α_j	β_j	γ_j	$LR^{+\dagger}$
1	SIGNOR	LIT	620,651	2,090	0.38 [0.36, 0.40]	5.9×10^{-4} [5.2×10^{-4}, 6.7×10^{-4}]	0.023 [0.017, 0.029]	562 [498, 635]
2	TRRUST	LIT	1,323,309	7,051	0.67 [0.65, 0.69]	9.7×10^{-4} [9.0×10^{-4}, 0.0010]	0.12 [0.10, 0.13]	472 [443, 504]
3	HTRIdb	LIT	419,212	2,231	0.41 [0.39, 0.44]	0.0015 [0.0014, 0.0017]	0.032 [0.025, 0.041]	250 [226, 278]
4	GOA	LIT	196,423	605	0.19 [0.17, 0.21]	0.0011 [9.5×10^{-4}, 0.0013]	0.029 [0.021, 0.038]	155 [127, 191]
5	FANTOM4	LIT	26,992	370	0.51 [0.45, 0.57]	0.0032 [0.0025, 0.0039]	0.12 [0.095, 0.15]	144 [114, 182]
6	NTNU.Curated	LIT	1,598,852	8,992	0.31 [0.29, 0.33]	1.2×10^{-4} [8.5×10^{-5}, 1.5×10^{-4}]	0.55 [0.54, 0.57]	139 [128, 149]
7	KEGG	LIT	95,997	1,223	0.48 [0.45, 0.51]	0.0041 [0.0036, 0.0046]	0.032 [0.023, 0.042]	115 [101, 131]
8	GEREDB	LIT	1,548,971	7,382	0.28 [0.27, 0.29]	0.0018 [0.0018, 0.0019]	0.22 [0.21, 0.23]	105 [97.7, 111]
9	TFactS	LIT	317,586	2,892	0.39 [0.37, 0.41]	0.0036 [0.0033, 0.0038]	0.080 [0.070, 0.090]	101 [93.0, 110]
10	ExTRI	LIT	4,139,536	25,919	0.46 [0.44, 0.48]	0.0012 [0.0011, 0.0013]	0.96 [0.94, 0.97]	94.8 [89.7, 100]
11	CytReg	LIT	61,210	1,202	0.51 [0.47, 0.55]	0.0072 [0.0065, 0.0079]	0.24 [0.21, 0.27]	64.1 [56.5, 72.0]
12	TRED	LIT	369,700	6,626	0.66 [0.63, 0.68]	0.0097 [0.0094, 0.010]	0.15 [0.13, 0.16]	63.9 [61.1, 67.1]
13	TFe	LIT	18,811	464	0.55 [0.49, 0.61]	0.0083 [0.0070, 0.0098]	0.17 [0.13, 0.21]	62.0 [50.3, 75.0]
14	Pavlidis2021	LIT	162,059	1,298	0.24 [0.22, 0.27]	0.0036 [0.0033, 0.0039]	0.10 [0.090, 0.12]	60.9 [53.2, 69.7]
15	IntAct	LIT	76,221	729	0.25 [0.22, 0.28]	0.0045 [0.0040, 0.0051]	0.049 [0.038, 0.061]	52.2 [44.2, 61.4]
16	ORegAnno	LIT	29,187	464	0.25 [0.21, 0.29]	0.012 [0.010, 0.013]	0.023 [0.013, 0.039]	21.3 [17.2, 26.0]
17	PAZAR	LIT	58,761	1,289	0.33 [0.28, 0.38]	0.018 [0.017, 0.019]	0.055 [0.035, 0.077]	17.7 [15.1, 20.8]
18	Other literature	LIT	1,715	247	0.62 [0.53, 0.70]	0.079 [0.066, 0.096]	0.17 [0.10, 0.25]	7.83 [6.21, 9.71]
19	JASPAR	MOTIF	14,214	790	0.080 [0.072, 0.089]	0.016 [0.0076, 0.026]	0.79 [0.58, 0.99]	1.66 [1.40, 1.96]
20	HOCOMOCO	MOTIF	15,200	862	0.079 [0.071, 0.088]	0.023 [0.014, 0.030]	0.66 [0.49, 0.92]	1.59 [1.34, 1.87]
21	ARACNe-GTEx	COEXP	15,952	1,392	0.11 [0.099, 0.12]	0.075 [0.069, 0.081]	–	1.43 [1.26, 1.62]

Observed: edges the annotator observes; reports: edges it reports. γ_j is the reporting rate under the burst class of the annotator's own family (no burst class for the single coexpression annotator, –).

[†] Post hoc: defined after the held-out data had been examined.

Table S4: Label-free model comparison: 5-fold cross-validated held-out log-likelihood per observed edge (folds formed by TF), and the difference to Family-HB per edge and in total (nats).

Configuration	Held-out log-lik. per edge (SE)	Difference per edge (SE)	Total (nats)	Folds better
Global universe				
Data-driven families	-0.0681 (0.0034)	2.0×10^{-5} (1.2×10^{-5})	100.7	4/5
Family-HB	-0.0681 (0.0034)	0 (0)	0.0	0/5
Lineage families	-0.0682 (0.0034)	-4.3×10^{-5} (2.7×10^{-5})	-220.0	1/5
Multifactor	-0.0682 (0.0033)	-8.2×10^{-5} (2.1×10^{-5})	-417.5	0/5
Dependence-aware	-0.0683 (0.0034)	-1.6×10^{-4} (2.8×10^{-5})	-790.7	0/5
EM baseline	-0.0687 (0.0034)	-6.3×10^{-4} (5.7×10^{-5})	-3224.5	0/5
Family-regularized	-0.0687 (0.0034)	-6.3×10^{-4} (5.7×10^{-5})	-3225.0	0/5
Singleton families	-0.0687 (0.0034)	-6.3×10^{-4} (5.7×10^{-5})	-3225.1	0/5
Immune universe				
Family-HB	-0.1395 (0.0154)	0 (0)	0.0	0/5
Dependence-aware	-0.1396 (0.0153)	-1.0×10^{-4} (2.2×10^{-4})	-12.0	1/5
Multifactor	-0.1396 (0.0153)	-1.1×10^{-4} (2.0×10^{-4})	-12.1	3/5
Lineage families	-0.1396 (0.0154)	-1.2×10^{-4} (1.5×10^{-4})	-14.4	2/5
Data-driven families	-0.1396 (0.0155)	-1.5×10^{-4} (1.1×10^{-4})	-19.7	1/5
Family-regularized	-0.1414 (0.0155)	-0.0019 (5.2×10^{-4})	-244.9	0/5
Singleton families	-0.1414 (0.0155)	-0.0019 (5.2×10^{-4})	-245.9	0/5
EM baseline	-0.1414 (0.0155)	-0.0019 (5.2×10^{-4})	-246.0	0/5

Standard errors in parentheses (over folds). Folds better: folds in which the configuration has a higher held-out log-likelihood than Family-HB.

Table S5: Posterior predictive checks: largest absolute z-score (observed minus replicated mean, divided by the replicated standard deviation) over the elements of each statistic, and the replicated share of edges with a literature report among the edges observed by the most literature annotators (top stratum), against the observed share.

	Reports per edge		Per annotator		Co-reporting log OR		Top stratum
Estimator	count	any report	rates	cond. count	within	between	replicated
Global universe; top stratum ($\geq$8 literature annotators) observed 0.14							
EM baseline	192.2	167.4	47.8	25.6	41.8	2.8	0.031
Dependence-aware	113.1	208.2	44.1	25.8	45.0	3.4	0.032
Multifactor	124.3	138.6	61.4	25.8	40.1	2.9	0.032
Family-regularized	240.3	226.6	55.3	25.6	59.1	3.8	0.031
Family-HB	121.2	179.8	51.1	25.1	22.5	4.2	0.031
Singleton families	150.9	223.3	47.0	25.6	41.9	2.6	0.032
Lineage families	166.3	238.1	55.5	25.2	27.1	3.9	0.032
Data-driven families	145.5	184.8	50.8	25.1	21.2	3.1	0.031
Immune universe; top stratum ($\geq$8 literature annotators) observed 0.16							
EM baseline	43.9	41.4	13.5	5.7	10.2	5.5	0.040
Dependence-aware	25.2	45.0	14.5	6.0	9.9	6.8	0.041
Multifactor	37.1	40.5	13.1	6.0	7.9	5.9	0.041
Family-regularized	49.8	44.6	12.8	5.6	9.9	6.8	0.041
Family-HB	38.7	34.9	16.0	5.9	6.6	4.4	0.039
Singleton families	50.7	36.9	13.9	5.7	6.5	6.8	0.039
Lineage families	31.2	65.3	26.5	5.9	5.4	8.6	0.041
Data-driven families	49.7	35.7	14.9	5.9	9.4	5.9	0.041

Reporting count: share of edges with k reporting annotators; any report by coverage: share of edges with at least one report by number of observing annotators; reporting rates and conditional count: per annotator; co-reporting: pairwise log odds ratios of reports within and between families. Replicates per fit: 20; the $|z|$ values carry Monte Carlo error from these few replicates (fits with nearly identical parameters can differ markedly), so the replicated rates are the better summary.

Table S6: Simulation-based recovery at the fitted Family-HB parameters (data simulated from the fitted model and refitted; design-faithful simulator primary, two simulators of the observation design as sensitivity analyses): parameter recovery (mean over annotators), ranking and calibration of credibility, and the realised false discovery rate of the Strong set; Family-HB also by the posterior mode its multi-start MAP selected (truth start: Family-HB started at the true parameters).

	w_T	α	β	Credibility			Strong set†		
Estimator	bias	RMSE	RMSE	AUROC	Spearman	ECE†	claimed	realised	n
Design-faithful (primary), global universe: 5 replicates, competing mode in 0†									
Oracle posterior	–	–	–	0.866	1.000	0.001	0.050	0.052	1,537
Family-HB, truth start	-5.6×10^{-5}	0.071	0.0039	0.866	0.992	0.006	0.050	0.057	1,572
Family-HB (MAP)	-5.6×10^{-5}	0.071	0.0039	0.866	0.992	0.006	0.050	0.057	1,572
truth mode ($n = 5$)	-5.6×10^{-5}	0.071	0.0039	–	–	–	–	0.057	1,572
EM baseline	0.0036	0.184	0.0094	0.856	0.819	0.405	0.005	0.575	8,337
Design-faithful (primary), immune universe: 20 replicates, competing mode in 2†									
Oracle posterior	–	–	–	0.911	1.000	0.003	0.049	0.052	142
Family-HB, truth start	-2.9×10^{-5}	0.112	0.0067	0.911	0.994	0.013	0.049	0.062	144
Family-HB (MAP)	3.6×10^{-4}	0.145	0.0076	0.907	0.974	0.066	0.049	0.149	151
truth mode ($n = 18$)	-3.3×10^{-5}	0.113	0.0068	–	–	–	–	0.060	146
competing mode ($n = 2$)	0.0038	0.309	0.013	–	–	–	–	0.944	199
EM baseline	0.0094	0.197	0.0097	0.900	0.867	0.469	0.030	0.621	895
Stratum-specific weights (sensitivity), global universe: 5 replicates, competing mode in 0†									
Oracle posterior	–	–	–	0.904	1.000	0.001	0.050	0.049	4,509
Family-HB, truth start	0.0018	0.053	0.0028	0.894	0.851	0.039	0.048	0.065	4,244
Family-HB (MAP)	0.0018	0.053	0.0028	0.894	0.851	0.039	0.048	0.065	4,244
truth mode ($n = 5$)	0.0018	0.053	0.0028	–	–	–	–	0.065	4,244
EM baseline	0.0063	0.184	0.0098	0.896	0.828	0.468	0.048	0.643	22,962
Stratum-specific weights (sensitivity), immune universe: 20 replicates, competing mode in 0†									
Oracle posterior	–	–	–	0.948	1.000	0.004	0.050	0.051	380
Family-HB, truth start	0.0025	0.075	0.0059	0.940	0.893	0.053	0.049	0.105	364
Family-HB (MAP)	0.0025	0.075	0.0059	0.940	0.893	0.053	0.049	0.105	364
truth mode ($n = 20$)	0.0025	0.075	0.0059	–	–	–	–	0.105	364
EM baseline	0.013	0.189	0.010	0.936	0.771	0.491	0.049	0.626	1,512
Fixed observation masks (sensitivity), global universe: 5 replicates, competing mode in 1									
Oracle posterior	–	–	–	0.866	1.000	0.002	0.050	0.045	1,471
Family-HB (MAP)	1.5×10^{-4}	0.148	0.0040	0.861	0.947	0.080	0.049	0.227	1,416
truth mode ($n = 4$)	-4.0×10^{-5}	0.043	0.0042	–	–	–	–	0.046	1,436
competing mode ($n = 1$)	9.3×10^{-4}	0.320	0.0030	–	–	–	–	0.953	1,334
EM baseline	0.0035	0.177	0.0097	0.856	0.819	0.402	0.005	0.574	8,223
Fixed observation masks (sensitivity), immune universe: 20 replicates, competing mode in 4									
Oracle posterior	–	–	–	0.911	1.000	0.005	0.049	0.055	138
Family-HB (MAP)	3.4×10^{-4}	0.192	0.0051	0.901	0.952	0.111	0.048	0.230	155
truth mode ($n = 16$)	-4.1×10^{-4}	0.143	0.0052	–	–	–	–	0.055	135
competing mode ($n = 4$)	0.0034	0.320	0.0049	–	–	–	–	0.932	235
EM baseline	0.0087	0.197	0.0097	0.901	0.866	0.454	0.017	0.600	765

Bias of w_T and root-mean-square error (RMSE) of the annotator rates α and β on the probability scale (oracle: true parameters, –). ECE: expected calibration error over credibility bins 0.1–0.9, pooled over replicates. Strong set: BFDR ≤ 0.05; claimed: its posterior expected FDR; realised: share of simulated non-edges in it. Competing mode: replicates whose selected optimum is not the truth-aligned one; the by-mode rows have few replicates. Design-faithful: real literature observation masks, motif and coexpression masks rebuilt from the simulated literature reports as in the evidence builder. Its data contain 59% (immune) and 62% (global) of the real number of motif- or coexpression-observed edges, i.e. fewer informative observations than the real data. Stratum-specific: the same with class weights that depend on the literature-mask stratum, which reproduces the real literature report counts. Fixed masks: every observation mask fixed at the real one, so that motif and coexpression observations do not depend on the simulated literature reports.

† Post hoc: defined after the held-out data had been examined.

Table S7: Mode-selection simulation: data simulated from the dominant Family-HB optimum (truth = dominant mode) or, in the immune universe, from the second optimum (truth = second mode), refitted by multi-start MAP; how often the highest-posterior optimum is the truth, how the gap between the two optima compares with the real gaps, and what the Strong set of each optimum realises. Rows by simulator of the observation design.[†]

	Global	Immune	Immune, truth = second mode
Design-faithful simulator (primary)			
Replicates	20	100	30
P(dominant optimum = truth)	1.00 [0.84, 1.00]	0.93 [0.86, 0.97]	0.60 [0.42, 0.75]
P(truth optimum found)	1.00 [0.84, 1.00]	1.00 [0.96, 1.00]	1.00 [0.89, 1.00]
Median gap between the two optima (nats)	49.8 [32.6, 63.4]	4.6 [1.2, 8.3]	1.6 [0.2, 3.8]
Share of simulated gaps below the real gap[g]	1.00	0.84	1.00
Mean Laplace probability of the other optimum	4.7×10^{-14}	0.12	0.24
Strong realised FDR, dominant optimum	0.054 [0.036, 0.064]	0.125 [0.025, 0.140]	0.415 [0.042, 0.962]
P(realised FDR $>$ 0.10), dominant	0.00	0.22	0.43
Strong realised FDR, truth-start	0.054 [0.036, 0.064]	0.065 [0.025, 0.118]	0.067 [0.036, 0.096]
Strong realised FDR, other optimum	0.945 [0.930, 0.961]	0.869 [0.857, 0.967]	0.593 [0.055, 0.960]
Strong realised FDR, Laplace mixture	0.054 [0.036, 0.064]	0.051 [0.000, 0.111] n = 67	0.295 [0.000, 0.990] n = 13
Strong realised FDR, oracle	0.051 [0.041, 0.058]	0.050 [0.030, 0.072]	0.052 [0.036, 0.075]
Strong realised FDR, EM	0.572 [0.563, 0.581]	0.611 [0.565, 0.665]	0.434 [0.398, 0.473]
P(truth ahead by $\geq$ 7.5 nats, immune)	1.00 [0.84, 1.00]	0.16 [0.10, 0.24]	0.00 [0.00, 0.11]
P(other ahead by $\geq$ 7.5 nats, immune)	0.00 [0.00, 0.16]	0.00 [0.00, 0.04]	0.00 [0.00, 0.11]
P(dominant = truth $\mid$ gap $\geq$ 7.5, immune)	1.00 [0.84, 1.00] n = 20	1.00 [0.81, 1.00] n = 16	–
Logistic P(dominant = truth) at 7.5, immune	–	1.00 [0.99, 1.00]	(0.79)[e]
P(dominant = truth $\mid$ gap $\geq$ 7.9, kidney)	1.00 [0.84, 1.00] n = 20	1.00 [0.78, 1.00] n = 14	–
Logistic P(dominant = truth) at 7.9, kidney	–	1.00 [0.99, 1.00]	(0.80)[e]
P(dominant = truth $\mid$ gap $\geq$ 11.6, lung)	1.00 [0.84, 1.00] n = 20	1.00 [0.34, 1.00] n = 2	–

Table S7: Mode-selection simulation: data simulated from the dominant Family-HB optimum (truth = dominant mode) or, in the immune universe, from the second optimum (truth = second mode), refitted by multi-start MAP; how often the highest-posterior optimum is the truth, how the gap between the two optima compares with the real gaps, and what the Strong set of each optimum realises. Rows by simulator of the observation design.[†] (continued)

	Global	Immune	Immune, truth = second mode
Design-faithful, stratum-specific class weights (sensitivity)			
Replicates	20	100	30
P(dominant optimum = truth)	1.00 [0.84, 1.00]	1.00 [0.96, 1.00]	0.97 [0.83, 0.99]
Median gap between the two optima (nats)	101.1 [93.4, 117.0]	9.6 [5.6, 15.4]	3.3 [1.0, 8.9]
Share of simulated gaps below the real gap[g]	0.60	0.31	0.87
Strong realised FDR, dominant optimum	0.061 [0.049, 0.079]	0.095 [0.060, 0.136]	0.097 [0.048, 0.103]
Strong realised FDR, other optimum	0.941 [0.937, 0.947]	0.954 [0.936, 0.972]	0.928 [0.935, 0.975]
Strong realised FDR, oracle	0.049 [0.046, 0.054]	0.051 [0.037, 0.065]	0.050 [0.041, 0.060]
P(truth ahead by $\geq$ 7.5 nats, immune)	1.00 [0.84, 1.00]	0.69 [0.59, 0.77]	0.13 [0.05, 0.30]
P(other ahead by $\geq$ 7.5 nats, immune)	0.00 [0.00, 0.16]	0.00 [0.00, 0.04]	0.00 [0.00, 0.11]
P(dominant = truth $\mid$ gap $\geq$ 108.3, global)	1.00 [0.68, 1.00] n = 8	–	–
P(dominant = truth $\mid$ gap $\geq$ 7.5, immune)	1.00 [0.84, 1.00] n = 20	1.00 [0.95, 1.00] n = 69	1.00 [0.51, 1.00] n = 4
P(dominant = truth $\mid$ gap $\geq$ 7.9, kidney)	1.00 [0.84, 1.00] n = 20	1.00 [0.94, 1.00] n = 66	1.00 [0.51, 1.00] n = 4
P(dominant = truth $\mid$ gap $\geq$ 11.6, lung)	1.00 [0.84, 1.00] n = 20	1.00 [0.88, 1.00] n = 29	–

Table S7: Mode-selection simulation: data simulated from the dominant Family-HB optimum (truth = dominant mode) or, in the immune universe, from the second optimum (truth = second mode), refitted by multi-start MAP; how often the highest-posterior optimum is the truth, how the gap between the two optima compares with the real gaps, and what the Strong set of each optimum realises. Rows by simulator of the observation design.† (continued)

	Global	Immune	Immune, truth = second mode
Fixed observation masks (sensitivity)			
Replicates	10	100	30
P(dominant optimum = truth)	0.70 [0.40, 0.89]	0.68 [0.58, 0.76]	0.77 [0.59, 0.88]
Median gap between the two optima (nats)	3.7 [2.4, 6.0]	0.8 [0.1, 3.4]	1.1 [0.3, 2.5]
Share of simulated gaps below the real gap[g]	1.00	0.98	1.00
Strong realised FDR, dominant optimum	0.319 [0.046, 0.939]	0.338 [0.026, 0.952]	0.279 [0.058, 0.951]
Strong realised FDR, other optimum	0.683 [0.052, 0.962]	0.656 [0.035, 0.967] $n = 99$	0.740 [0.068, 0.974]
Strong realised FDR, oracle	0.050 [0.045, 0.059]	0.051 [0.028, 0.073]	0.053 [0.034, 0.069]
P(truth ahead by ≥ 7.5 nats, immune)	0.10 [0.02, 0.40]	0.01 [0.00, 0.05]	0.00 [0.00, 0.11]
P(other ahead by ≥ 7.5 nats, immune)	0.00 [0.00, 0.28]	0.00 [0.00, 0.04]	0.00 [0.00, 0.11]
P(dominant = truth $\mid$ gap ≥ 7.5, immune)	1.00 [0.21, 1.00] $n = 1$	0.50 [0.09, 0.91] $n = 2$	–
P(dominant = truth $\mid$ gap ≥ 7.9, kidney)	–	0.00 [0.00, 0.79] $n = 1$	–
P(dominant = truth $\mid$ gap ≥ 11.6, lung)	–	0.00 [0.00, 0.79] $n = 1$	–

Design-faithful: real literature observation masks, motif and coexpression masks rebuilt from the simulated literature reports as in the evidence builder. Its data contain 59% (immune) and 62% (global) of the real number of motif- or coexpression-observed edges, i.e. fewer informative observations than the real data. Stratum-specific: the same with class weights that depend on the literature-mask stratum, which reproduces the real literature report counts. Fixed masks: every observation mask fixed at the real one, so that motif and coexpression observations do not depend on the simulated literature reports.

[g] Real gap of the simulated universe: 108.3 nats (global), 7.5 nats (immune); the immune columns use the immune gap.

Laplace mixture: replicates whose mixture Strong set is empty are omitted (n: replicates used; immune, 33 of 100 omitted, among them 7 of the 7 replicates in which the wrong optimum is dominant; dominant optimum on the replicates used 0.062; truth = second mode, 17 of 30 omitted, among them 8 of the 12 replicates in which the wrong optimum is dominant; dominant optimum on the replicates used 0.342): the mixture abstains rather than errs in the omitted replicates, so its realised FDR is not comparable with the dominant-optimum row over all replicates.

Intervals: probabilities, Wilson 95% interval (logistic rows: Wald 95%); medians and means, 10th–90th percentile over replicates. Gaps: real log-posterior gaps of the fitted universes (global, immune, kidney, lung). Truth-start: fit started at the true parameters; other optimum: the next-best optimum of the same fit; Laplace mixture: credibility averaged over the optima with Laplace weights.

[e] Extrapolated (flagged in the source table): too few replicates reach that gap, so the logistic curve is evaluated beyond the simulated gaps (no interval); rows in which every cell is extrapolated are omitted.

† Post hoc: defined after the held-out data had been examined.

Table S8: Governance under a two-mode mixture posterior: credibility averaged over the dominant and the second Family-HB mode with second-mode weight p_2 (bridge-sampling estimate, its upper Monte-Carlo bound, the Laplace estimate, or fixed values), tier sizes, Jaccard index with the primary sets, and the posterior expected FDR of the primary dual Strong and promoted sets under the mixture.[†]

	p_2		Tier size (dual)			Jaccard		FDR of primary	
Scenario	ctx.	glob.	ctx. Strong	Strong	prom.	Strong	prom.	Strong	prom.
Immune									
dominant mode (primary)	0	0	334	297	493	1.00	1.00	0.036	0.159
dominant mode, re-run	0	0	333	297	493	1.00	1.00	0.036	0.159
bridge p_2	0.0044	1.8×10^{-46}	326	296	492	1.00	1.00	0.040	0.162
bridge, upper bound	0.0069	1.8×10^{-46}	321	292	492	0.98	1.00	0.042	0.164
Laplace p_2	0.0016	3.9×10^{-46}	331	297	493	1.00	1.00	0.037	0.160
fixed p_2	0.01	1.8×10^{-46}	315	291	490	0.98	0.99	0.045	0.166
fixed p_2, both fits	0.01	0.01	315	273	484	0.92	0.98	0.045	0.166
fixed p_2	0.05	1.8×10^{-46}	31	31	480	0.10	0.97	0.082	0.195
fixed p_2, both fits	0.05	0.05	31	27	449	0.09	0.91	0.082	0.195
fixed p_2	0.10	1.8×10^{-46}	0	0	441	0.00	0.89	0.129	0.231
fixed p_2, both fits	0.10	0.10	0	0	388	0.00	0.79	0.129	0.231
second mode only	1.00	1.8×10^{-46}	438	0	18	0.00	0.02	0.967	0.880
Kidney									
dominant mode (primary)	0	0	179	149	276	1.00	1.00	0.034	0.157
dominant mode, re-run	0	0	179	149	276	1.00	1.00	0.034	0.158
bridge p_2	0.0029	1.8×10^{-46}	176	149	276	1.00	1.00	0.037	0.159
bridge, upper bound	0.0048	1.8×10^{-46}	173	148	275	0.99	1.00	0.038	0.161
Laplace p_2	0.0073	3.9×10^{-46}	170	148	275	0.99	1.00	0.041	0.162
fixed p_2	0.01	1.8×10^{-46}	166	146	274	0.98	0.99	0.043	0.164
fixed p_2, both fits	0.01	0.01	166	138	269	0.93	0.97	0.043	0.164
fixed p_2	0.05	1.8×10^{-46}	36	36	261	0.24	0.95	0.081	0.194
fixed p_2, both fits	0.05	0.05	36	17	244	0.11	0.88	0.081	0.194
fixed p_2	0.10	1.8×10^{-46}	0	0	242	0.00	0.88	0.128	0.231
fixed p_2, both fits	0.10	0.10	0	0	206	0.00	0.75	0.128	0.231
second mode only	1.00	1.8×10^{-46}	183	0	11	0.00	0.03	0.971	0.897
Lung									
dominant mode (primary)	0	0	232	204	412	1.00	1.00	0.041	0.167
dominant mode, re-run	0	0	233	205	412	1.00	1.00	0.041	0.166
bridge p_2	6.3×10^{-5}	1.8×10^{-46}	232	204	412	1.00	1.00	0.041	0.167
bridge, upper bound	1.0×10^{-4}	1.8×10^{-46}	232	204	412	1.00	1.00	0.041	0.168
Laplace p_2	8.9×10^{-5}	3.9×10^{-46}	232	204	412	1.00	1.00	0.041	0.168
fixed p_2	0.01	1.8×10^{-46}	210	193	411	0.95	1.00	0.051	0.175
fixed p_2, both fits	0.01	0.01	210	187	409	0.92	0.99	0.051	0.175
fixed p_2	0.05	1.8×10^{-46}	26	26	391	0.13	0.95	0.088	0.205
fixed p_2, both fits	0.05	0.05	26	22	368	0.11	0.89	0.088	0.205
fixed p_2	0.10	1.8×10^{-46}	0	0	337	0.00	0.82	0.136	0.244
fixed p_2, both fits	0.10	0.10	0	0	306	0.00	0.74	0.136	0.244
second mode only	1.00	1.8×10^{-46}	188	0	1	0.00	0.00	0.986	0.928

p_2: weight of the second mode in the context and the global fit (the global fit keeps its own estimate unless marked "both fits"). Ctx. Strong: single-fit Strong set of the context fit; Strong and prom.: dual Strong and promoted sets. FDR: posterior expected FDR of the primary sets under the mixture credibility of the context fit.

[†] Post hoc: defined after the held-out data had been examined.

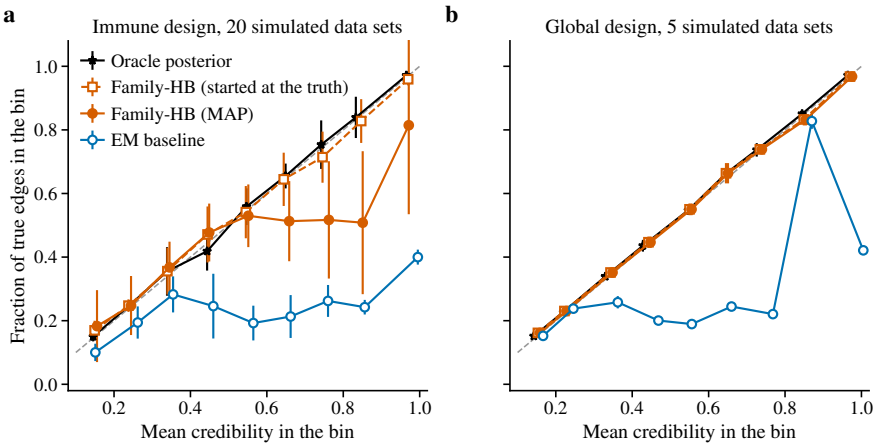


Figure S2: Calibration of credibility in simulation. Data simulated from the fitted Family-HB model with the real literature observation masks and the motif/coexpression masks rebuilt from the simulated literature reports (design-faithful simulator; the simulated data contain 59% (immune) and 62% (global) of the real number of motif- or coexpression-observed edges). Edges are binned by credibility (bins from 0.1 to 1; the bin below 0.1, which holds almost all edges, is omitted); points: pooled fraction of truly regulatory edges against the mean credibility in the bin; bars: ± 1 SD of that fraction across simulated data sets; dashed line: perfect calibration. Started at the true parameters, Family-HB is close to the oracle posterior; multi-start MAP selects the competing mode in some data sets (immune 2 of 20, global 0 of 5), which pulls the upper bins down and widens their spread; the EM baseline is over-confident (fraction of true edges more than 0.01 below the mean credibility) in 16 of its 18 universe $\times$ bin cells. The design-faithful simulation is post hoc (defined after the held-out data had been examined; it uses no held-out data).

S3 Sensitivity analyses and tuning on evaluation data

Fits under alternative priors, evidence variants and family definitions, their held-out performance, and the experiment that tunes configurations on perturbation labels to measure the optimism of such tuning (Section 5.9).

Table S9: Sensitivity of the Family-HB fit (posterior mode) to the evidence variant, the family grouping and the priors, against the primary fit: class weight w_T, Spearman correlation of credibilities over observed edges, size and Jaccard index of the posterior-mode BFDR ≤ 0.05 set, and the posterior-mode structure (distinct optima among the restarts, gap to the second optimum and its Laplace probability).

Fit	w_T	Spearman	BFDR set n	Jaccard	Optima	Gap†	P(2nd)†
Global universe (primary: w_T = 0.0031, BFDR set 4,064 edges)							
Literal observation mask	0.0038	0.939	5,609	0.60	3	2794.9	$< 10^{-300}$ [u]
Both endpoints required	0.0036	0.820	6,127	0.58	4	3144.9	$< 10^{-300}$ [u]
Both endpoints, curated-cond.	0.0028	0.872	4,270	0.89	2	137.7	2.4×10^{-58}
ChIP family included	0.0033	0.975	4,561	0.88	4	294.9	1.5×10^{-128}
ExTRI and NTNU merged	0.0027	0.949	2,569	0.62	3	144.6	2.6×10^{-61}
Without very large regulons	0.0030	0.945	3,890	0.90	3	101.3	5.1×10^{-43}
Without ChIP-derived LIT	0.0028	0.986	3,349	0.77	3	121.0	6.8×10^{-52}
Literature only[a]	0.0031	1.000	3,978	0.93	1	–	–
Singleton families	0.0068	0.816	18,873	0.21	1	–	–
Lineage families	0.0044	0.890	8,929	0.44	2	24.3	4.0×10^{-9}
Data-driven families	0.0031	1.000	4,065	0.99	2	250.3	1.5×10^{-108}
Scale priors halved	0.0031	1.000	4,060	1.00	2	109.8	5.2×10^{-47}
Scale priors doubled	0.0031	1.000	4,066	1.00	2	107.2	2.1×10^{-45}
Dirichlet(0.5)	0.0031	1.000	4,056	1.00	2	108.0	5.4×10^{-46}
Dirichlet(2)	0.0031	1.000	4,071	1.00	2	109.1	2.0×10^{-46}
Immune universe (primary: w_T = 0.0063, BFDR set 331 edges)							
Literal observation mask	0.0073	0.954	476	0.69	2	181.9	1.9×10^{-80}
Both endpoints required	0.0069	0.893	471	0.70	2	192.7	4.4×10^{-85}
Both endpoints, curated-cond.	0.0054	0.913	317	0.94	2	10.9	9.2×10^{-5}
ChIP family included	0.0065	0.984	350	0.95	3	8.9	1.4×10^{-4}
ExTRI and NTNU merged	0.0063	0.969	290	0.84	2	4.5	0.026
Without very large regulons	0.0064	0.985	335	0.99	2	8.1	8.3×10^{-4}
Without ChIP-derived LIT	0.0066	0.991	321	0.83	2	7.6	4.4×10^{-4}
Literature only[a]	0.0063	0.999	323	0.96	1	–	–
Singleton families	0.015	0.859	1,284	0.26	1	–	–
Lineage families	0.0086	0.953	593	0.53	1	–	–
Data-driven families	0.0063	1.000	331	1.00	2	7.5	0.0016
Scale priors halved	0.0062	1.000	327	0.99	2	7.7	0.0052
Scale priors doubled	0.0063	1.000	334	0.99	2	6.8	0.0033
Dirichlet(0.5)	0.0063	1.000	331	1.00	2	7.2	0.020
Dirichlet(2)	0.0063	1.000	332	1.00	2	7.5	0.0011

[a] With one modality family the true and literature-burst classes are exchangeable; the labelling is not identified. Gap: log-posterior units; – where all restarts reached one optimum. BFDR set: posterior-mode set (the Strong tier of the governance tables uses NUTS credibility).

[u] Below the smallest double-precision number (the Laplace mass underflows at gaps of thousands of nats).

[†] Post hoc: defined after the held-out data had been examined.

Table S10: Held-out AP lift of the sensitivity fits and its paired difference to the primary Family-HB (MAP) fit (95% TF-cluster bootstrap interval), all evaluable edges.

Fit	ENCODE TFBS	ENETS2	Pooled perturbation
Global universe			
Family-HB (MAP), primary fit: AP lift	0.96 [0.94, 0.98]	1.59 [1.31, 2.00]	0.94 [0.90, 0.98]
Family-HB (NUTS)	0.000 [0.000, 0.000]	0.000 [−0.002, 0.002]	0.000 [−0.0002, 0.000]
Family-HB (NUTS, random init.)^m	0.005 [0.002, 0.009]	0.095 [−0.009, 0.233]	0.005 [−0.006, 0.014]
Literal observation mask	0.017 [0.004, 0.033]	0.790 [0.545, 1.106]	0.020 [−0.021, 0.088]
Both endpoints required	−0.031 [−0.048, −0.013]	0.721 [0.491, 1.023]	0.053 [−0.002, 0.122]
Both endpoints, curated-cond.	−0.009 [−0.023, 0.004]	0.022 [−0.007, 0.057]	−0.002 [−0.035, 0.027]
ChIP family included	0.051 [0.036, 0.069]	0.549 [0.402, 0.756]	0.029 [−0.0002, 0.076]
ExTRI and NTNU merged	0.000 [−0.002, 0.001]	0.081 [0.033, 0.151]	0.006 [0.002, 0.010]
Without very large regulons	0.019 [0.012, 0.025]	−0.063 [−0.139, −0.001]	0.004 [−0.010, 0.016]
Without ChIP-derived LIT	0.010 [0.006, 0.014]	−0.034 [−0.106, 0.038]	0.008 [−0.003, 0.017]
Literature only	0.000 [−0.0004, 0.000]	−0.056 [−0.096, −0.026]	−0.001 [−0.003, −0.0002]
Singleton families	0.006 [0.001, 0.011]	−0.091 [−0.242, 0.019]	0.004 [−0.012, 0.018]
Lineage families	0.005 [0.002, 0.008]	−0.049 [−0.138, 0.024]	0.002 [−0.009, 0.011]
Data-driven families	0.000 [−0.0003, −0.000]	−0.011 [−0.024, −0.0004]	0.000 [−0.0001, 0.001]
Scale priors halved	0.000 [0.000, 0.000]	0.002 [0.001, 0.005]	0.000 [−0.0001, 0.000]
Scale priors doubled	0.000 [−0.000, −0.000]	−0.001 [−0.002, −0.0002]	0.000 [−0.0001, 0.000]
Dirichlet(0.5)	0.000 [−0.000, −0.000]	−0.001 [−0.001, −0.0001]	0.000 [−0.000, 0.000]
Dirichlet(2)	0.000 [−0.000, 0.000]	0.002 [0.001, 0.004]	0.000 [−0.0001, 0.000]

Table S10: Held-out AP lift of the sensitivity fits and its paired difference to the primary Family-HB (MAP) fit (95% TF-cluster bootstrap interval), all evaluable edges. (continued)

Fit	ENCODE TFBS	ENETS2	Pooled perturbation
Immune universe			
Family-HB (MAP), primary fit: AP lift	0.97 [0.95, 1.00]	1.61 [1.15, 2.77]	0.94 [0.88, 1.01]
Family-HB (NUTS)	0.001 [−0.0001, 0.002]	−0.010 [−0.033, 0.001]	0.000 [−0.001, 0.003]
Family-HB (NUTS, random init.)[m]	0.007 [−0.002, 0.018]	−0.160 [−0.825, 0.121]	−0.016 [−0.035, −0.001]
Literal observation mask	0.050 [0.018, 0.097]	0.446 [0.077, 0.916]	−0.022 [−0.062, 0.033]
Both endpoints required	−0.008 [−0.042, 0.039]	0.329 [−0.082, 0.788]	0.017 [−0.031, 0.079]
Both endpoints, curated-cond.	0.001 [−0.007, 0.009]	−0.123 [−0.296, 0.027]	0.028 [0.000, 0.062]
ChIP family included	0.089 [0.052, 0.146]	0.264 [0.174, 0.428]	−0.018 [−0.039, 0.024]
ExTRI and NTNU merged	−0.001 [−0.003, 0.002]	−0.049 [−0.269, 0.160]	0.006 [−0.0001, 0.012]
Without very large regulons	0.005 [0.002, 0.008]	−0.072 [−0.216, 0.027]	−0.004 [−0.008, 0.001]
Without ChIP-derived LIT	0.001 [−0.002, 0.006]	−0.178 [−0.610, 0.021]	0.003 [−0.004, 0.011]
Literature only	−0.001 [−0.003, 0.000]	−0.009 [−0.061, 0.018]	0.000 [−0.003, 0.002]
Singleton families	0.007 [−0.0005, 0.018]	−0.219 [−0.910, 0.059]	−0.010 [−0.035, 0.009]
Lineage families	0.005 [0.000, 0.011]	−0.152 [−0.745, 0.154]	−0.004 [−0.019, 0.006]
Data-driven families	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Scale priors halved	0.000 [−0.0001, 0.000]	0.003 [−0.002, 0.025]	0.000 [−0.0004, 0.002]
Scale priors doubled	0.000 [−0.0001, 0.001]	0.002 [−0.002, 0.016]	0.000 [−0.002, 0.000]
Dirichlet(0.5)	0.000 [−0.000, 0.000]	0.001 [−0.0001, 0.002]	0.000 [−0.0001, 0.000]
Dirichlet(2)	0.000 [−0.000, 0.000]	0.000 [−0.002, 0.001]	0.000 [−0.000, 0.000]
Scale priors halved (NUTS)[m]	0.008 [−0.001, 0.020]	−0.159 [−0.824, 0.126]	−0.016 [−0.034, −0.001]
Scale priors doubled (NUTS)[m]	0.007 [−0.002, 0.018]	−0.137 [−0.820, 0.162]	−0.016 [−0.035, −0.001]

First row: AP lift of the primary Family-HB (MAP) fit with its 95% TF-cluster bootstrap interval, as in the main held-out table. Other rows: AP lift of the fit minus AP lift of the primary fit on the same evaluable edges (paired TF-cluster bootstrap of its own).

[m] NUTS from random starts whose chains mix posterior modes (maximum $\widehat{R}$ 1.556–1.742): the difference reflects mode mixing, not the prior or the initialisation.

Table S11: Optimism of tuning the configuration on perturbation labels (pooled perturbation reference, global universe): configuration selected on one half of the TFs (or all but one study) and scored on the other, against the label-free choice (best cross-validated log-likelihood) and the fixed primary Family-HB.

Selection	Selected (share)	Distinct	In	Out	Optimism	Out − prim.	*P*
all configurations (19 configurations), random TF halves (50 splits)							
Criterion: pooled AP lift							
tuned on labels	Family-HB, both endpoints required (0.74)	4	0.994	0.979	0.015	0.040	0.94
label-free (CV)	Data-driven families (1.00)	1	0.937	0.940	−0.003	0.000	0.86
primary	Family-HB (1.00)	1	0.937	0.940	−0.003	0.000	0.00
Criterion: per-TF AP lift (geom. mean)							
tuned on labels†	Family-HB, both endpoints required (0.96)	2	1.241	1.288	−0.047	0.097	1.00
label-free (CV)†	Data-driven families (1.00)	1	1.152	1.196	−0.045	0.005	0.62
primary†	Family-HB (1.00)	1	1.148	1.191	−0.044	0.000	0.00
all configurations (19 configurations), leave one study out (4 splits)							
Criterion: pooled AP lift							
tuned on labels	Family-HB, both endpoints required (0.75)	2	1.007	1.020	−0.013	0.016	0.50
label-free (CV)	Data-driven families (1.00)	1	0.953	1.005	−0.052	0.001	1.00
primary	Family-HB (1.00)	1	0.953	1.004	−0.051	0.000	0.00
Criterion: per-TF AP lift (geom. mean)							
tuned on labels†	Family-HB, both endpoints required (1.00)	1	1.271	1.440	−0.168	0.123	1.00
label-free (CV)†	Data-driven families (1.00)	1	1.177	1.317	−0.140	0.000	0.75
primary†	Family-HB (1.00)	1	1.174	1.317	−0.143	0.000	0.00
primary evidence only (12 configurations), random TF halves (50 splits)							
Criterion: pooled AP lift							
tuned on labels	Data-driven families (0.30)	9	0.942	0.939	0.003	−0.001	0.54
label-free (CV)	Data-driven families (1.00)	1	0.937	0.940	−0.003	0.000	0.86
primary	Family-HB (1.00)	1	0.937	0.940	−0.003	0.000	0.00
Criterion: per-TF AP lift (geom. mean)							
tuned on labels†	Lineage families (0.60)	6	1.162	1.191	−0.029	0.000	0.60
label-free (CV)†	Data-driven families (1.00)	1	1.152	1.196	−0.045	0.005	0.62
primary†	Family-HB (1.00)	1	1.148	1.191	−0.044	0.000	0.00
primary evidence only (12 configurations), leave one study out (4 splits)							
Criterion: pooled AP lift							
tuned on labels	Singleton families (0.25)	4	0.958	1.061	−0.103	0.056	1.00
label-free (CV)	Data-driven families (1.00)	1	0.953	1.005	−0.052	0.001	1.00
primary	Family-HB (1.00)	1	0.953	1.004	−0.051	0.000	0.00
Criterion: per-TF AP lift (geom. mean)							
tuned on labels†	Lineage families (0.75)	2	1.188	1.319	−0.131	0.002	0.75
label-free (CV)†	Data-driven families (1.00)	1	1.177	1.317	−0.140	0.000	0.75
primary†	Family-HB (1.00)	1	1.174	1.317	−0.143	0.000	0.00

Selected: most frequently selected configuration (share of splits); distinct: number of different configurations selected. In / out: criterion on the tuning / held-out part; optimism: in − out; out − primary: held-out gain over Family-HB; *P*: share of splits with a held-out gain.

† Post hoc: defined after the held-out data had been examined.

Table S12: Perturbation positives and credibility by evidence status for every tuning configuration (pooled perturbation reference, global universe): observed over TF-matched expected positives and mean credibility of reported, observed but unreported, and unobserved edges.[†]

Configuration	Reported		Observed, not reported		Unobserved	
	O/E	$\bar{c}$	O/E	$\bar{c}$	O/E	$\bar{c}$
EM baseline	1.36	0.54	1.01	0.0023	0.97	0.0068
Dependence-aware	1.36	0.24	1.01	5.7×10^{-4}	0.97	0.0024
Multifactor	1.36	0.029	1.01	4.5×10^{-5}	0.97	2.2×10^{-4}
Family-regularized	1.36	0.54	1.01	0.0023	0.97	0.0068
Family-HB	1.36	0.24	1.01	0.0011	0.97	0.0031
Family-HB, literal observation mask	1.28	0.23	1.00	0.0017	0.96	0.0038
Family-HB, both endpoints required	1.28	0.25	1.00	0.0015	0.97	0.0036
Family-HB, both endpoints, curated-cond.	1.36	0.26	1.00	0.0010	0.99	0.0028
Family-HB, chIP family included	1.05	0.044	1.00	0.0015	0.97	0.0033
Family-HB, exTRI and NTNU merged	1.36	0.21	1.01	9.5×10^{-4}	0.97	0.0027
Family-HB, without very large regulons	1.36	0.24	1.02	0.0011	0.95	0.0030
Family-HB, without ChIP-derived LIT	1.36	0.22	1.02	9.4×10^{-4}	0.96	0.0028
Scale priors halved	1.36	0.24	1.01	0.0011	0.97	0.0031
Scale priors doubled	1.36	0.25	1.01	0.0011	0.97	0.0031
Dirichlet(0.5)	1.36	0.24	1.01	0.0011	0.97	0.0031
Dirichlet(2)	1.36	0.25	1.01	0.0011	0.97	0.0031
Singleton families	1.36	0.54	1.01	0.0023	0.97	0.0068
Lineage families	1.36	0.36	1.01	0.0015	0.97	0.0044
Data-driven families	1.36	0.25	1.01	0.0011	0.97	0.0031

Positive rate over all evaluable edges: 0.097. The status classes of the evidence variants differ because their observation rules differ.

[†] Post hoc: defined after the held-out data had been examined.

Table S13: A joint selection objective that weighs dropout robustness against the perturbation reward, $J = w_s \mathrm{Rob}_{\mathrm{source}} + w_b \mathrm{Rob}_{\mathrm{block}} + w_p \mathrm{Pert}$ over the weight simplex: share of weight points at which each candidate is selected, for two robustness metrics and two candidate sets, with the inputs of the objective.

	Objective inputs			Share of weight points selected			
Candidate	D source	D block	Pert	F_1, LC	F_1, all	prec., LC	prec., all
EM baseline	0.008	0.220	0.939	0.18	0.00	0.02	0.00
Dependence-aware	0.017	0.220	0.937	0.00	0.00	0.00	0.00
Multifactor	0.058	0.309	0.932	0.00	0.00	0.00	0.00
Family-regularized	0.009	0.218	0.939	0.82	0.00	0.62	0.00
Family-HB	0.017	0.227	0.935	0.00	0.00	0.36	0.00
Union	0.005	0.037	1.013	–	0.92	–	0.91
Majority ($\geq$2)	0.014	0.073	1.011	–	0.00	–	0.00
Vote fraction	0.007	0.059	1.014	–	0.02	–	0.05
Fisher	0.022	0.100	1.015	–	0.00	–	0.00
Stouffer	0.019	0.081	1.017	–	0.06	–	0.05

D: mean rank disagreement under source and block (family and lineage) dropout (F_1 metric); Pert: pooled perturbation AP lift. Candidate sets: the five latent-class estimators (LC) or these and the five combination rules (all). Weight points: 66.

S4 Held-out evaluation: further results

All scores against all held-out references in the global universe, the universe decomposition for the context fits, and the evaluation of the foundation-model and expression-based predicted sets in the tissue universes, with their uniform-attention nulls and the sensitivity of the predicted sets (Sections 5.3, 5.5 and 5.6).

Table S14: Held-out evaluation in the global universe, all scores (all evaluable edges): ChIP-seq references.

	ENCODE TFBS			ENCODE-strict			ENETS2		
Score	AP lift	AUROC	Enr@500	AP lift	AUROC	Enr@500	AP lift	AUROC	Enr@500
Fam.-HB NUTS	0.96 [0.94, 0.98]	0.481	1.33 [1.16, 1.68]	0.97 [0.94, 1.00]	0.491	1.47 [1.19, 2.12]	1.58 [1.31, 2.00]	0.522	6.66 [5.06, 11.56]
Fam.-HB	0.96 [0.94, 0.98]	0.481	1.32 [1.16, 1.67]	0.97 [0.94, 1.00]	0.491	1.47 [1.20, 2.12]	1.59 [1.31, 2.00]	0.522	6.79 [5.09, 11.56]
EM	0.96 [0.95, 0.98]	0.489	1.33 [1.19, 1.59]	0.97 [0.95, 1.00]	0.500	1.49 [1.24, 1.93]	1.49 [1.26, 1.83]	0.529	7.09 [4.81, 10.84]
Dep.	0.96 [0.95, 0.98]	0.485	1.39 [1.19, 1.70]	0.97 [0.95, 1.01]	0.496	1.41 [1.18, 2.09]	1.59 [1.31, 2.02]	0.525	7.07 [5.48, 10.78]
Multif.	0.96 [0.94, 0.97]	0.482	1.45 [1.25, 1.80]	0.96 [0.94, 0.99]	0.492	1.50 [1.25, 2.17]	1.70 [1.40, 2.15]	0.516	12.28 [8.40, 18.08]
Fam.-reg.	0.96 [0.95, 0.98]	0.489	1.33 [1.19, 1.59]	0.97 [0.95, 1.00]	0.500	1.49 [1.24, 1.92]	1.50 [1.26, 1.83]	0.529	7.05 [4.79, 10.89]
Union	1.01 [1.00, 1.01]	0.505	1.13 [1.08, 1.20]	1.01 [1.00, 1.03]	0.511	1.16 [1.09, 1.33]	1.33 [1.19, 1.53]	0.533	2.82 [2.23, 3.83]
Majority	1.00 [1.00, 1.01]	0.504	1.20 [1.12, 1.34]	1.01 [1.00, 1.02]	0.507	1.22 [1.10, 1.47]	1.30 [1.17, 1.49]	0.523	3.90 [2.92, 5.48]
Votes	1.01 [1.00, 1.01]	0.505	1.46 [1.28, 1.76]	1.01 [1.00, 1.04]	0.511	1.63 [1.30, 2.25]	1.60 [1.34, 1.97]	0.533	8.39 [6.13, 11.90]
Vote frac.	1.01 [1.00, 1.01]	0.505	1.09 [1.03, 1.17]	1.01 [1.00, 1.04]	0.511	1.14 [1.06, 1.28]	1.47 [1.26, 1.79]	0.533	2.50 [1.55, 3.83]
Families	1.01 [1.00, 1.01]	0.505	1.44 [1.28, 1.71]	1.01 [1.00, 1.04]	0.511	1.62 [1.28, 2.48]	1.62 [1.38, 1.98]	0.533	8.90 [6.09, 13.41]
Fisher	1.07 [1.05, 1.08]	0.547	1.50 [1.29, 1.81]	1.10 [1.07, 1.15]	0.557	1.61 [1.31, 2.23]	1.80 [1.48, 2.27]	0.537	9.34 [6.60, 13.81]
Stouffer	1.04 [1.03, 1.06]	0.534	1.53 [1.33, 1.85]	1.07 [1.04, 1.11]	0.542	1.65 [1.32, 2.27]	1.80 [1.47, 2.28]	0.527	9.70 [7.09, 14.97]
DoRothEA	1.01 [1.00, 1.01]	0.504	1.18 [1.09, 1.41]	1.01 [1.00, 1.03]	0.508	1.20 [1.08, 1.66]	1.37 [1.20, 1.62]	0.523	4.10 [3.02, 7.26]
CollecTRI	1.00 [1.00, 1.01]	0.504	1.48 [1.23, 2.04]	1.01 [1.00, 1.02]	0.508	1.49 [1.21, 2.12]	1.43 [1.25, 1.70]	0.528	6.59 [4.38, 8.91]
Status†	1.05 [1.03, 1.07]	0.537	1.13 [1.08, 1.20]	1.06 [1.04, 1.10]	0.543	1.16 [1.09, 1.33]	1.38 [1.24, 1.59]	0.540	2.82 [2.23, 3.83]
Fam.-HB unobs. last†	1.07 [1.05, 1.10]	0.545	1.32 [1.16, 1.67]	1.11 [1.06, 1.18]	0.555	1.47 [1.20, 2.12]	1.79 [1.46, 2.25]	0.537	6.79 [5.09, 11.56]
Random	1.00 [1.00, 1.00]	0.500	1.01 [0.94, 1.08]	1.00 [1.00, 1.01]	0.499	0.91 [0.84, 0.98]	1.01 [0.99, 1.03]	0.503	1.11 [0.41, 1.92]

AP lift: 95% TF-cluster bootstrap interval; AUROC: within-TF AUROC; Enr@500: stratified enrichment of the top 500 edges.
† Post hoc: defined after the held-out data had been examined.

Table S15: Held-out evaluation in the global universe, all scores (all evaluable edges): perturbation cohorts, AP lift and within-TF AUROC.

	Norman		Papalexi*		Replogle		Pooled perturbation	
Score	AP lift	AUROC	AP lift	AUROC	AP lift	AUROC	AP lift	AUROC
Fam.-HB NUTS	0.94 [0.91, 0.98]	0.469	0.96 [0.84, 2.40]	0.553	0.98 [0.89, 1.15]	0.481	0.94 [0.90, 0.98]	0.478
Fam.-HB	0.94 [0.91, 0.98]	0.469	0.96 [0.84, 2.39]	0.553	0.98 [0.89, 1.15]	0.481	0.94 [0.90, 0.98]	0.478
EM	0.95 [0.93, 0.98]	0.469	0.97 [0.86, 1.96]	0.553	0.99 [0.90, 1.15]	0.480	0.94 [0.90, 0.98]	0.479
Dep.	0.95 [0.92, 0.98]	0.468	0.96 [0.85, 2.28]	0.550	0.98 [0.89, 1.14]	0.481	0.94 [0.90, 0.98]	0.478
Multif.	0.94 [0.91, 0.98]	0.470	0.97 [0.85, 2.54]	0.485	0.98 [0.89, 1.15]	0.478	0.93 [0.89, 0.98]	0.473
Fam.-reg.	0.95 [0.93, 0.98]	0.469	0.97 [0.86, 1.96]	0.553	0.99 [0.90, 1.15]	0.480	0.94 [0.90, 0.98]	0.479
Union	1.01 [1.00, 1.03]	0.502	1.05 [0.99, 1.89]	0.532	1.00 [1.00, 1.02]	0.512	1.01 [1.00, 1.03]	0.507
Majority	1.01 [1.00, 1.02]	0.502	1.05 [1.00, 1.57]	0.521	1.01 [1.00, 1.03]	0.511	1.01 [1.00, 1.03]	0.507
Votes	1.02 [1.01, 1.04]	0.502	1.08 [1.00, 2.01]	0.532	1.01 [1.00, 1.05]	0.512	1.02 [1.01, 1.05]	0.507
Vote frac.	1.01 [1.00, 1.02]	0.502	1.06 [1.00, 2.34]	0.533	1.01 [1.00, 1.03]	0.512	1.01 [1.00, 1.03]	0.507
Families	1.02 [1.01, 1.03]	0.502	1.10 [1.00, 2.26]	0.532	1.01 [1.00, 1.03]	0.512	1.02 [1.00, 1.05]	0.507
Fisher	0.99 [0.97, 1.02]	0.484	1.08 [0.93, 2.30]	0.542	1.04 [0.99, 1.13]	0.507	1.01 [0.98, 1.06]	0.497
Stouffer	0.99 [0.97, 1.01]	0.490	1.10 [0.94, 2.36]	0.523	1.04 [0.99, 1.13]	0.512	1.02 [0.99, 1.06]	0.502
DoRothEA	1.01 [1.00, 1.01]	0.501	1.03 [1.00, 1.68]	0.520	1.00 [1.00, 1.02]	0.509	1.01 [1.00, 1.02]	0.505
CollecTRI	1.02 [1.01, 1.03]	0.502	1.06 [1.00, 1.59]	0.526	1.01 [1.00, 1.06]	0.512	1.02 [1.01, 1.04]	0.507
Status†	1.05 [1.02, 1.09]	0.511	1.11 [1.01, 1.96]	0.544	1.03 [0.93, 1.12]	0.526	1.06 [1.02, 1.12]	0.519
Fam.-HB unobs. last†	1.00 [0.97, 1.04]	0.486	1.08 [0.92, 2.62]	0.576	1.03 [0.93, 1.19]	0.509	1.03 [0.99, 1.09]	0.501
Random	0.99 [0.98, 1.00]	0.495	1.02 [0.99, 1.09]	0.531	1.00 [0.98, 1.02]	0.505	0.99 [0.99, 1.00]	0.508

* Low support (fewer than 10 TFs with positives or fewer than 20 positives): Papalexi.
Not shown: Dixit d7, Dixit d13 (almost no positives at the fold-change threshold, Table 5); their results are in the analysis tables of the code repository.
AP lift: 95% TF-cluster bootstrap interval; AUROC: within-TF AUROC; Enr@500: stratified enrichment of the top 500 edges.
† Post hoc: defined after the held-out data had been examined.

Table S16: Held-out evaluation in the global universe, all scores (all evaluable edges): perturbation cohorts, stratified enrichment of the top 500 edges.

	Norman	Papalexi*	Replogle	Pooled perturbation
Score	Enr@500	Enr@500	Enr@500	Enr@500
Fam.-HB NUTS	1.31 [1.06, 1.58]	1.93 [1.35, 3.30]	1.33 [1.06, 2.82]	1.53 [1.21, 1.98]
Fam.-HB	1.31 [1.06, 1.58]	1.95 [1.34, 3.30]	1.33 [1.06, 2.82]	1.53 [1.21, 1.98]
EM	1.24 [1.01, 1.50]	1.72 [1.25, 3.18]	1.39 [1.14, 3.24]	1.47 [1.19, 1.94]
Dep.	1.32 [1.08, 1.58]	2.01 [1.35, 3.27]	1.34 [1.10, 2.93]	1.55 [1.24, 1.94]
Multif.	1.32 [1.10, 1.54]	1.76 [1.30, 4.39]	1.61 [1.16, 2.71]	1.77 [1.32, 2.32]
Fam.-reg.	1.24 [1.01, 1.50]	1.72 [1.25, 3.18]	1.39 [1.14, 3.24]	1.45 [1.19, 1.95]
Union	1.15 [1.01, 1.33]	1.45 [1.25, 2.74]	1.30 [1.08, 2.33]	1.25 [1.11, 1.41]
Majority	1.28 [1.05, 1.51]	1.67 [1.40, 3.45]	1.40 [1.16, 3.66]	1.41 [1.18, 1.68]
Votes	1.29 [1.08, 1.52]	1.88 [1.44, 3.72]	1.43 [1.12, 3.89]	1.59 [1.19, 2.15]
Vote frac.	1.23 [1.05, 1.48]	1.80 [1.42, 3.92]	1.39 [1.03, 2.56]	1.42 [1.12, 1.85]
Families	1.26 [1.09, 1.49]	1.80 [1.44, 3.90]	1.42 [1.15, 2.83]	1.84 [1.33, 2.39]
Fisher	1.26 [1.07, 1.51]	1.84 [1.43, 3.43]	1.42 [1.15, 3.04]	1.52 [1.19, 2.14]
Stouffer	1.28 [1.07, 1.50]	1.78 [1.42, 3.62]	1.44 [1.17, 3.02]	1.55 [1.19, 2.13]
DoRothEA	1.20 [1.04, 1.40]	1.87 [1.34, 3.76]	1.39 [1.03, 3.08]	1.43 [1.17, 1.97]
CollecTRI	1.33 [1.09, 1.57]	1.94 [1.40, 3.38]	1.47 [1.18, 3.28]	1.68 [1.20, 2.27]
Status†	1.15 [1.01, 1.33]	1.45 [1.25, 2.74]	1.30 [1.08, 2.33]	1.25 [1.11, 1.41]
Fam.-HB unobs. last†	1.31 [1.06, 1.58]	1.95 [1.34, 3.30]	1.33 [1.06, 2.82]	1.53 [1.21, 1.98]
Random	0.94 [0.76, 1.11]	0.93 [0.68, 1.29]	0.93 [0.69, 1.18]	0.85 [0.69, 1.03]

* Low support (fewer than 10 TFs with positives or fewer than 20 positives): Papalexi.
Not shown: Dixit d7, Dixit d13 (almost no positives at the fold-change threshold, Table 5); their results are in the analysis tables of the code repository.
AP lift: 95% TF-cluster bootstrap interval; AUROC: within-TF AUROC; Enr@500: stratified enrichment of the top 500 edges.
† Post hoc: defined after the held-out data had been examined.

Table S17: Universe-aware decomposition of the context–global difference in raw precision of the top 500 edges (context-fit Family-HB credibility): raw ratio = base-rate ratio × enrichment ratio, and comparison with prevalence-matched random sub-universes of the global universe drawn edge by edge or as whole TF blocks.

	Base rate		P@500			Edge null		TF-block null†	
Tissue	Context	Ratio	Context	Raw ratio	Enrichment ratio	Median	Surviving	Median	Surviving
ENCODE TFBS (global universe: P@500 = 0.64, base rate 0.38)									
Immune	0.24	0.62	0.42	0.65 [0.47, 0.87]	1.04 [0.79, 1.36]	0.35 [0.31, 0.40]	−0.27	0.32 [0.21, 0.48]	−0.42
Kidney	0.36	0.95	0.42	0.65 [0.52, 0.85]	0.68 [0.57, 0.90]	0.48 [0.43, 0.52]	0.24	0.43 [0.33, 0.60]	0.04
Lung	0.20	0.52	0.40	0.60 [0.41, 0.86]	1.15 [0.91, 1.54]	0.28 [0.25, 0.32]	−0.42	0.25 [0.16, 0.40]	−0.55
Pooled perturbation (global universe: P@500 = 0.19, base rate 0.10)									
Immune	0.25	2.57	0.26	1.34 [0.63, 2.48]	0.52 [0.29, 0.78]	0.24 [0.23, 0.27]	0.24	0.26 [0.22, 0.31]	0.08
Kidney	0.15	1.55	0.14	0.61 [0.28, 1.39]	0.39 [0.26, 0.74]	0.15 [0.14, 0.17]	0.11	0.15 [0.12, 0.20]	0.12
Lung	0.29	3.04	0.29	1.37 [0.72, 2.47]	0.45 [0.28, 0.72]	0.29 [0.27, 0.31]	0.00	0.30 [0.27, 0.35]	−0.16

Ratios are context over global. Null: median and 95% interval of P@500 over random sub-universes with the context's number of evaluable edges and exactly its number of positives, drawn as edges or as whole TF blocks matched to the context's per-TF sizes and positive counts. Surviving: fraction of the context–global gap not reproduced by the null, (context − null median)/(context − global); negative values mean that the context value exceeds its null although raw precision falls. The edge-level null ignores the clustering of edges within TFs and is anti-conservative; the TF-block null keeps it.
Intervals: TF-cluster bootstrap of the context and the global set drawn independently (conservative when the sets share TFs)†.
† Post hoc: defined after the held-out data had been examined.

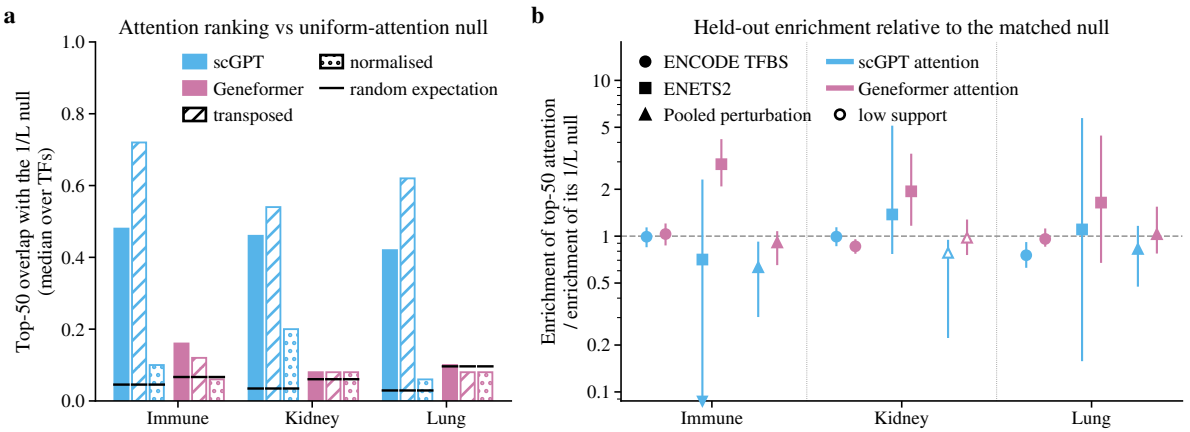


Figure S3: Foundation-model attention against its uniform-attention null. (a) Median over TFs of the overlap between the 50 top-ranked targets by attention and by the $1/L$ null (the attention an edge would receive if each cell spread attention uniformly over its L input genes); filled: raw attention, hatched: transposed and normalised variants; black ticks: overlap expected for a random ranking. The 50 top targets of raw scGPT attention overlap with those of its null by 0.42–0.48 (random ranking 0.03–0.05), those of raw Geneformer attention by 0.08–0.16. (b) TF-matched held-out enrichment of each model's top-50-per-TF predicted set divided by that of its own $1/L$ null, with paired 95% TF-cluster bootstrap intervals (reference line at 1; triangles mark lower limits below the axis range; open markers: references flagged as low support).

Table S18: Network-inference methods in the tissue-context universes: TF-matched enrichment of each method's predicted set (top 50 targets per TF) among held-out positives, and its ratio to the method's matched null (the $1/L$ uniform-attention null of the same model for attention scores, otherwise the coverage indicator); 95% TF-cluster bootstrap intervals.

	ENCODE TFBS		ENETS2		Pooled perturbation	
Method	Enrichment	vs null†	Enrichment	vs null†	Enrichment	vs null†
Immune						
scGPT	0.62 [0.48, 0.77]	0.99 [0.85, 1.14]	0.46 [0.00, 1.04]	0.71 [0.00, 2.31]	0.76 [0.36, 1.17]	0.63 [0.30, 0.92]
scGPT norm.	0.84 [0.73, 0.97]	1.35 [1.18, 1.59]	0.73 [0.41, 1.04]	1.12 [0.47, 7.74]	0.68 [0.39, 0.96]	0.56 [0.33, 0.78]
scGPT 1/L	0.62 [0.51, 0.73]	0.56 [0.46, 0.64]	0.65 [0.00, 1.36]	0.98 [0.00, 2.24]	1.21 [0.93, 1.47]	1.26 [0.97, 1.65]
Geneformer	1.40 [1.19, 1.66]	1.03 [0.87, 1.21]	4.77 [2.43, 7.42]	2.90 [2.09, 4.20]	1.13 [0.88, 1.27]	0.91 [0.65, 1.08]
Geneformer norm.	1.42 [1.22, 1.66]	1.05 [0.89, 1.23]	4.50 [2.33, 6.93]	2.73 [1.96, 4.05]	1.19 [0.96, 1.35]	0.96 [0.69, 1.17]
Geneformer 1/L	1.36 [1.16, 1.63]	0.83 [0.73, 0.95]	1.65 [0.84, 2.69]	0.85 [0.56, 1.15]	1.24 [1.07, 1.46]	1.52 [1.17, 2.14]
GRNBoost2	1.74 [1.53, 2.01]	1.40 [1.19, 1.64]	5.83 [3.73, 8.68]	4.17 [2.10, 8.17]	1.28 [1.14, 1.44]	1.24 [1.03, 1.56]
GENIE3	1.19 [0.99, 1.44]	1.11 [0.91, 1.38]	3.88 [2.21, 5.81]	8.58 [3.28, 31.06]	1.25 [1.06, 1.37]	1.23 [1.03, 1.42]
|Pearson|	1.69 [1.48, 1.96]	1.67 [1.43, 2.00]	5.68 [3.25, 8.75]	6.75 [2.72, 29.28]	1.26 [1.14, 1.38]	1.18 [1.00, 1.34]
|Spearman|	1.64 [1.43, 1.91]	1.62 [1.38, 1.93]	5.27 [3.02, 7.96]	6.26 [2.60, 25.99]	1.26 [1.06, 1.44]	1.17 [0.95, 1.38]
Fam.-HB NUTS	1.30 [1.16, 1.47]	–	3.83 [2.41, 5.50]	–	1.29 [0.96, 1.64]	–
Random	0.92 [0.84, 0.99]	–	0.79 [0.36, 1.20]	–	1.06 [0.84, 1.28]	–
Kidney						
scGPT	0.55 [0.48, 0.61]	0.99 [0.86, 1.14]	1.21 [0.69, 1.93]	1.38 [0.77, 5.12]	0.75* [0.24, 0.86]	0.78 [0.22, 0.95]
scGPT norm.	0.65 [0.58, 0.71]	1.17 [1.00, 1.38]	1.07 [0.50, 1.94]	1.21 [0.49, 5.30]	0.67* [0.00, 0.79]	0.70 [0.00, 0.89]
scGPT 1/L	0.55 [0.47, 0.64]	0.51 [0.43, 0.61]	0.88 [0.20, 1.50]	1.11 [0.25, 2.08]	0.96* [0.73, 1.37]	1.11 [0.72, 1.51]
Geneformer	1.13 [1.00, 1.28]	0.86 [0.77, 0.96]	2.33 [1.48, 3.09]	1.94 [1.17, 3.38]	1.03* [0.84, 1.36]	0.98 [0.76, 1.28]
Geneformer norm.	1.12 [0.99, 1.27]	0.85 [0.77, 0.95]	2.24 [1.39, 3.01]	1.86 [1.07, 3.32]	1.04* [0.83, 1.43]	0.99 [0.74, 1.33]
Geneformer 1/L	1.31 [1.25, 1.38]	0.94 [0.89, 1.00]	1.20 [0.65, 1.82]	0.79 [0.44, 1.26]	1.05* [0.86, 1.33]	1.14 [0.97, 1.48]
GRNBoost2	1.20 [1.04, 1.36]	1.15 [0.95, 1.40]	2.42 [1.25, 3.50]	1.96 [0.99, 3.92]	1.16* [0.78, 1.52]	1.46 [0.93, 2.21]

Table S18: Network-inference methods in the tissue-context universes: TF-matched enrichment of each method's predicted set (top 50 targets per TF) among held-out positives, and its ratio to the method's matched null (the 1/*L* uniform-attention null of the same model for attention scores, otherwise the coverage indicator); 95% TF-cluster bootstrap intervals. (continued)

	ENCODE TFBS		ENETS2		Pooled perturbation	
Method	Enrichment	vs null†	Enrichment	vs null†	Enrichment	vs null†
|Pearson|	1.35 [1.18, 1.53]	1.39 [1.23, 1.58]	2.36 [0.82, 3.64]	2.35 [1.21, 4.46]	1.19* [0.78, 1.87]	1.42 [0.90, 2.36]
|Spearman|	1.32 [1.13, 1.52]	1.36 [1.19, 1.57]	2.52 [0.92, 3.92]	2.52 [1.19, 5.02]	1.17* [0.61, 2.20]	1.40 [0.75, 2.72]
Fam.-HB NUTS	1.00 [0.94, 1.08]	–	1.54 [1.00, 2.09]	–	1.11* [0.78, 1.43]	–
Random	1.04 [0.95, 1.13]	–	1.56 [0.69, 2.42]	–	1.04* [0.87, 1.24]	–
Lung						
scGPT	0.64 [0.49, 0.84]	0.75 [0.63, 0.92]	0.76 [0.16, 1.64]	1.10 [0.16, 5.73]	0.89 [0.55, 1.16]	0.83 [0.47, 1.16]
scGPT norm.	0.48 [0.36, 0.60]	0.57 [0.44, 0.73]	0.94 [0.33, 1.70]	1.36 [0.39, 5.68]	0.85 [0.69, 1.04]	0.79 [0.56, 1.05]
scGPT 1/L	0.85 [0.66, 1.09]	0.79 [0.59, 1.01]	0.69 [0.14, 1.42]	0.80 [0.17, 4.37]	1.07 [0.86, 1.42]	0.96 [0.73, 1.30]
Geneformer	1.53 [1.29, 1.88]	0.96 [0.86, 1.12]	1.58 [0.79, 2.59]	1.64 [0.67, 4.42]	0.94 [0.70, 1.28]	1.03 [0.77, 1.55]
Geneformer norm.	1.51 [1.26, 1.85]	0.94 [0.84, 1.10]	1.71 [0.92, 2.45]	1.77 [0.85, 4.40]	0.92 [0.68, 1.24]	1.00 [0.75, 1.52]
Geneformer 1/L	1.60 [1.35, 1.94]	1.02 [0.91, 1.17]	0.96 [0.45, 1.38]	0.68 [0.24, 1.40]	0.92 [0.66, 1.13]	1.20 [0.74, 1.83]
GRNBoost2	1.75 [1.45, 2.11]	1.48 [1.23, 1.90]	3.59 [2.40, 4.90]	4.82 [2.40, 21.18]	0.95 [0.65, 1.25]	0.89 [0.55, 1.29]
|Pearson|	1.66 [1.35, 2.12]	1.62 [1.22, 2.24]	2.63 [0.83, 3.84]	1.97 [0.42, 4.23]	0.96 [0.73, 1.13]	1.08 [0.67, 1.59]
|Spearman|	1.53 [1.20, 1.93]	1.49 [1.11, 2.06]	2.44 [1.03, 3.37]	1.82 [0.54, 4.14]	0.96 [0.74, 1.18]	1.08 [0.78, 1.42]
Fam.-HB NUTS	1.49 [1.30, 1.71]	–	3.68 [2.55, 5.29]	–	1.09 [0.87, 1.27]	–
Random	0.87 [0.70, 1.05]	–	0.88 [0.28, 1.41]	–	0.81 [0.61, 1.04]	–

* Low support (few evaluable TFs or positives).
Enrichment 1: as many positives as expected from the TF composition of the set. Ratio to null 1: no better than the matched null.
† Post hoc: defined after the held-out data had been examined.

Table S19: Sensitivity of the predicted sets in the tissue-context universes. (a) TF-matched enrichment among held-out ChIP positives of each method's predicted set of 20, 50 or 100 targets per TF (point estimates). (b) Attention restricted to pairs co-observed in at least 5 cells: share of the unrestricted top-50 set that is retained, and overlap of the top set with that of the uniform-attention ($1/L$) null.

(a) Predicted-set size

	ENCODE TFBS			ENETS2		
Method	$k = 20$	$k = 50$	$k = 100$	$k = 20$	$k = 50$	$k = 100$
Immune						
scGPT	0.65	0.62	0.69	0.24	0.46	0.75
Geneformer	1.40	1.40	1.42	5.09	4.77	4.45
GRNBoost2	1.89	1.74	1.63	6.65	5.83	4.28
|Pearson|	1.79	1.69	1.69	7.34	5.68	3.90
Fam.-HB NUTS	1.56	1.30	1.15	5.83	3.83	2.37
Random	0.89	0.92	0.96	0.59	0.79	1.13
Kidney						
scGPT	0.50	0.55	0.70	0.62	1.21	1.44
Geneformer	1.13	1.13	1.12	2.66	2.33	1.94
GRNBoost2	1.20	1.20	1.26	3.64	2.42	2.24
|Pearson|	1.37	1.35	1.37	2.98	2.36	2.11
Fam.-HB NUTS	1.15	1.00	0.94	2.83	1.54	1.16
Random	0.97	1.04	1.03	0.46	1.56	1.13
Lung						
scGPT	0.57	0.64	0.67	0.38	0.76	1.02
Geneformer	1.59	1.53	1.52	2.11	1.58	1.66
GRNBoost2	1.88	1.75	1.67	4.21	3.59	2.68
|Pearson|	1.77	1.66	1.62	3.05	2.63	2.45
Fam.-HB NUTS	1.98	1.49	1.25	6.24	3.68	2.38
Random	0.95	0.87	0.87	0.63	0.88	1.01

(b) Co-observation restriction

		Retained	Overlap with the $1/L$ null (%)		
Universe	Model	(%)	unrestricted	restricted	random
Immune	scGPT	5	48	41	5
Immune	Geneformer	31	16	16	11
Kidney	scGPT	18	46	22	4
Kidney	Geneformer	29	8	12	11
Lung	scGPT	6	42	35	4
Lung	Geneformer	22	10	20	17

Overlap: median over TFs of the share of a TF's top targets that are also among the top targets of the uniform-attention null; random: the overlap expected for a random ranking.

Table S20: Foundation-model attention against its uniform-attention (1/L) null in each tissue: coverage, Spearman correlation with the null, median overlap of the top-k targets per TF with the null's (and its expectation under random ranking), and the share of top-k pairs co-observed in only one cell or in at most four cells.

	Coverage	Spearman	Top-k overlap		Top-k co-observed in	
Score	(%)	with null	median	random	1 cell	$\leq$4 cells
Immune						
scGPT, raw	50.5	0.80	0.48	0.05	0.75	0.95
scGPT, raw, transposed	50.5	0.81	0.72	0.05	0.76	0.96
scGPT, normalised	50.5	0.02	0.10	0.05	0.67	0.89
scGPT, normalised, transposed	50.5	−0.06	0.02	0.05	0.59	0.79
scGPT, 1/L null	50.5	–	–	0.05	0.78	0.97
Geneformer, raw	20.5	0.11	0.16	0.07	0.46	0.69
Geneformer, raw, transposed	20.5	0.01	0.12	0.07	0.50	0.71
Geneformer, normalised	20.5	−0.09	0.06	0.07	0.44	0.67
Geneformer, normalised, transposed	20.5	−0.18	0.04	0.07	0.50	0.71
Geneformer, 1/L null	20.5	–	–	0.07	0.54	0.82
Kidney						
scGPT, raw	69.7	0.64	0.46	0.03	0.57	0.82
scGPT, raw, transposed	69.7	0.66	0.54	0.03	0.50	0.76
scGPT, normalised	69.7	−0.01	0.20	0.03	0.49	0.74
scGPT, normalised, transposed	69.7	−0.02	0.07	0.03	0.33	0.54
scGPT, 1/L null	69.7	–	–	0.03	0.61	0.89
Geneformer, raw	37.2	0.05	0.08	0.06	0.42	0.71
Geneformer, raw, transposed	37.2	−0.08	0.08	0.06	0.48	0.75
Geneformer, normalised	37.2	0.02	0.08	0.06	0.42	0.71
Geneformer, normalised, transposed	37.2	−0.11	0.06	0.06	0.48	0.75
Geneformer, 1/L null	37.2	–	–	0.06	0.42	0.79
Lung						
scGPT, raw	66.7	0.72	0.42	0.03	0.66	0.94
scGPT, raw, transposed	66.7	0.73	0.62	0.03	0.60	0.92
scGPT, normalised	66.7	−0.01	0.06	0.03	0.63	0.90
scGPT, normalised, transposed	66.7	−0.02	0.03	0.03	0.56	0.83
scGPT, 1/L null	66.7	–	–	0.03	0.63	0.93
Geneformer, raw	19.8	0.03	0.10	0.10	0.50	0.78
Geneformer, raw, transposed	19.8	−0.01	0.08	0.10	0.53	0.80
Geneformer, normalised	19.8	−0.04	0.08	0.10	0.49	0.77
Geneformer, normalised, transposed	19.8	−0.08	0.06	0.10	0.53	0.80
Geneformer, 1/L null	19.8	–	–	0.10	0.54	0.83

k = 50 targets per TF.

S5 Informative-observation extension

Fits, cross-validation and held-out evaluation of the post hoc selection-model extension of Section 3.4 (Section 5.4).

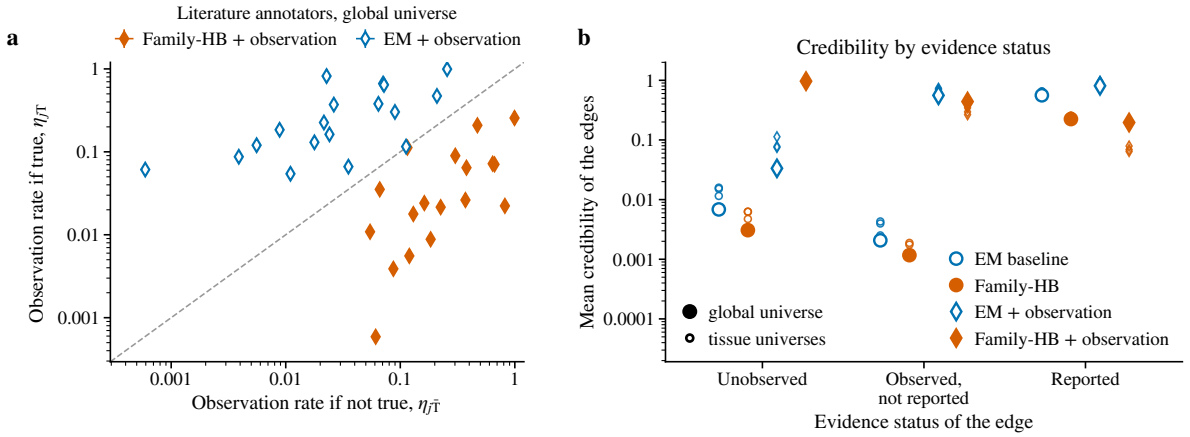


Figure S4: Informative-observation extension (secondary, post hoc; MAP fits with Laplace 95% intervals). (a) Observation rates of the 18 literature annotators under the true class and under the other classes at the best optimum of Family-HB + observation and EM + observation (global universe; dashed: equal rates, i.e. ignorable observation). The two models place most of their annotators on opposite sides of the diagonal (above it: 0 of 18 for Family-HB + observation, 18 of 18 for EM + observation): the label T is not anchored. (b) Mean credibility of unobserved, observed-but-unreported and reported edges for the ignorable models (EM, Family-HB) and the extensions (large markers: global universe; small open markers: immune, kidney and lung). In the global universe Family-HB + observation gives unobserved edges the highest mean credibility (0.97); EM + observation ranks them below observed-but-unreported edges (0.033 vs 0.56).

Table S21: Informative-observation extension (posterior mode, Laplace intervals) against the ignorable models: class weight of T, share of the joint log-likelihood due to observation, size of the BFDR ≤ 0.05 set, mean credibility by evidence status, literature annotators whose observation rate is higher under T, and agreement with the ignorable counterpart.

		Obs.	BFDR	Mean credibility			$\eta_T >$		Alt.
Fit	w_T	share	set n	unobs.	obs. neg.	reported	$\eta_{\bar{T}}$	Spearman	optimum
Global universe									
Family-HB	0.0031	–	4,064	0.0031	0.0012	0.22	–	–	–
+ observation†	0.68 [0.68, 0.68]	0.98	6,096,120	0.97	0.44	0.20	0/18	0.67	swapped
+ obs., TF-w.†	0.0031 [0.0030, 0.0033]	0.98	4,211	0.0031	0.0012	0.23	17/18	0.98	2nd optimum
EM baseline	0.0068	–	18,873	0.0068	0.0021	0.56	–	–	–
+ observation†	0.32 [0.32, 0.32]	0.98	2,275,210	0.033	0.56	0.81	18/18	−0.82	swapped
+ obs., TF-w.†	0.0068 [0.0067, 0.0070]	0.98	18,971	0.0068	0.0021	0.56	18/18	0.98	none
Immune universe									
Family-HB	0.0063	–	331	0.0063	0.0019	0.26	–	–	–
+ observation†	0.65 [0.65, 0.65]	0.97	169,386	0.89	0.30	0.064	0/18	0.76	swapped
+ obs., TF-w.†	0.0091 [0.0078, 0.011]	0.97	514	0.0089	0.0016	0.44	18/18	0.77	2nd optimum
EM baseline	0.015	–	1,281	0.015	0.0040	0.65	–	–	–
+ observation†	0.35 [0.35, 0.35]	0.96	89,098	0.11	0.70	0.94	18/18	−0.81	swapped
+ obs., TF-w.†	0.015 [0.014, 0.016]	0.97	1,283	0.014	0.0038	0.65	18/18	1.00	none
Kidney universe									
Family-HB	0.0063	–	189	0.0063	0.0018	0.25	–	–	–
+ observation†	0.68 [0.68, 0.68]	0.97	111,065	0.92	0.34	0.079	0/18	0.78	swapped
+ obs., TF-w.†	0.012 [0.0098, 0.014]	0.97	481	0.011	0.0027	0.53	18/18	0.83	2nd optimum
EM baseline	0.016	–	812	0.016	0.0043	0.65	–	–	–
+ observation†	0.32 [0.31, 0.32]	0.96	49,909	0.075	0.65	0.93	18/18	−0.82	swapped
+ obs., TF-w.†	0.015 [0.014, 0.017]	0.97	820	0.015	0.0040	0.66	18/18	0.99	none
Lung universe									
Family-HB	0.0047	–	245	0.0047	0.0011	0.25	–	–	–
+ observation†	0.63 [0.63, 0.63]	0.98	189,960	0.92	0.27	0.069	0/18	0.85	swapped
+ obs., TF-w.†	0.0073 [0.0063, 0.0086]	0.98	412	0.0072	0.0013	0.43	18/18	0.87	2nd optimum
EM baseline	0.012	–	1,230	0.012	0.0025	0.63	–	–	–
+ observation†	0.37 [0.37, 0.37]	0.97	113,129	0.078	0.73	0.93	17/18	−0.87	swapped
+ obs., TF-w.†	0.011 [0.010, 0.012]	0.98	1,231	0.011	0.0024	0.63	17/18	1.00	none

Obs. share: observation term over the joint log-likelihood. $\eta_T > \eta_{\bar{T}}$: literature annotators that observe true edges more often than non-edges. Spearman: credibility of observed edges against the ignorable counterpart. Alt. optimum: role of the next optimum (label-swapped: T and burst classes exchanged).

† Post hoc: defined after the held-out data had been examined.

Table S22: Cross-validated held-out log-likelihood (5 folds by TF) of the informative-observation models and their ignorable counterparts: of the reports given the observation pattern (per observed edge), of the observation indicators and of both (per edge).

Model	Reports | observation	Observation	Joint
Global universe			
Family-HB	-0.0681	$-2.4537^{\dagger}$	$-2.4909^{\dagger}$
Family-HB + obs.	-0.0684 $(-2.7 \times 10^{-4})^{\dagger}$	-2.0539 $(0.40)^{\dagger}$	-2.0913 $(0.40)^{\dagger}$
Family-HB + obs., TF-weighted	-0.0681 $(1.6 \times 10^{-5})^{\dagger}$	-2.4499 $(0.0039)^{\dagger}$	-2.4883 $(0.0026)^{\dagger}$
EM baseline	-0.0687	$-2.4537^{\dagger}$	$-2.4912^{\dagger}$
EM + obs.	-0.0847 $(-0.016)^{\dagger}$	-2.0539 $(0.40)^{\dagger}$	-2.1002 $(0.39)^{\dagger}$
EM + obs., TF-weighted	-0.0687 $(4.5 \times 10^{-5})^{\dagger}$	-2.4421 $(0.012)^{\dagger}$	-2.4849 $(0.0064)^{\dagger}$
Immune universe			
Family-HB	-0.1395	$-2.5555^{\dagger}$	$-2.6111^{\dagger}$
Family-HB + obs.	-0.1405 $(-1.0 \times 10^{-3})^{\dagger}$	-1.9097 $(0.65)^{\dagger}$	-1.9657 $(0.65)^{\dagger}$
Family-HB + obs., TF-weighted	-0.1392 $(3.0 \times 10^{-4})^{\dagger}$	-2.5312 $(0.024)^{\dagger}$	-2.6002 $(0.011)^{\dagger}$
EM baseline	-0.1414	$-2.5555^{\dagger}$	$-2.6118^{\dagger}$
EM + obs.	-0.1812 $(-0.040)^{\dagger}$	-1.9097 $(0.65)^{\dagger}$	-1.9819 $(0.63)^{\dagger}$
EM + obs., TF-weighted	-0.1407 $(6.8 \times 10^{-4})^{\dagger}$	-2.5220 $(0.033)^{\dagger}$	-2.5957 $(0.016)^{\dagger}$

In parentheses: difference to the ignorable counterpart (Family-HB or EM) on the same folds. For the ignorable models the observation term is a class-independent Bernoulli model of the literature observation indicators.
† Post hoc: defined after the held-out data had been examined.

Table S23: Held-out evaluation of the informative-observation extension in the global universe (all evaluable edges): AP lift [95% TF-cluster bootstrap interval], within-TF AUROC and stratified Enr@500, with the primary estimators, Fisher, the post hoc status scores and random.

Score	AP lift [95% CI]	Within-TF AUROC	Strat. Enr@500
ENCODE TFBS			
Family-HB + obs.†	0.97 [0.94, 0.99]	0.483	1.11
EM + obs.†	1.02 [1.01, 1.03]	0.517	1.07
Family-HB + obs., TF-weighted†	0.97 [0.94, 1.02]	0.481	1.32
EM + obs., TF-weighted†	0.98 [0.95, 1.02]	0.489	1.33
Family-HB + obs., swapped labels†	1.02 [1.01, 1.02]	0.516	1.05
EM + obs., swapped labels†	0.97 [0.94, 0.99]	0.484	1.11
Family-HB	0.96 [0.94, 0.98]	0.481	1.32
EM baseline	0.96 [0.95, 0.98]	0.489	1.33
Fisher	1.07 [1.05, 1.08]	0.547	1.50
3-level status†	1.05 [1.03, 1.07]	0.537	1.13
Family-HB, unobserved last†	1.07 [1.05, 1.10]	0.545	1.32
Random	1.00 [1.00, 1.00]	0.500	1.01
ENETS2			
Family-HB + obs.†	0.95 [0.91, 1.00]	0.474	1.31
EM + obs.†	1.22 [1.13, 1.34]	0.525	1.53
Family-HB + obs., TF-weighted†	1.75 [1.42, 2.19]	0.522	6.97
EM + obs., TF-weighted†	1.65 [1.38, 2.02]	0.529	7.07
Family-HB + obs., swapped labels†	1.14 [1.06, 1.22]	0.518	1.05
EM + obs., swapped labels†	0.95 [0.92, 1.00]	0.477	1.31
Family-HB	1.59 [1.31, 2.00]	0.522	6.79
EM baseline	1.49 [1.26, 1.83]	0.529	7.09
Fisher	1.80 [1.48, 2.27]	0.537	9.34
3-level status†	1.38 [1.24, 1.59]	0.540	2.82
Family-HB, unobserved last†	1.79 [1.46, 2.25]	0.537	6.79
Random	1.01 [0.99, 1.03]	0.503	1.11
Pooled perturbation			
Family-HB + obs.†	1.10 [0.99, 1.24]	0.463	0.99
EM + obs.†	1.13 [1.10, 1.18]	0.536	0.98
Family-HB + obs., TF-weighted†	0.93 [0.86, 1.05]	0.478	1.55
EM + obs., TF-weighted†	0.94 [0.87, 1.04]	0.479	1.46
Family-HB + obs., swapped labels†	1.12 [1.09, 1.17]	0.532	1.01
EM + obs., swapped labels†	1.10 [0.99, 1.24]	0.464	0.99
Family-HB	0.94 [0.90, 0.98]	0.478	1.53
EM baseline	0.94 [0.90, 0.98]	0.479	1.47
Fisher	1.01 [0.98, 1.06]	0.497	1.52
3-level status†	1.06 [1.02, 1.12]	0.519	1.25
Family-HB, unobserved last†	1.03 [0.99, 1.09]	0.501	1.53
Random	0.99 [0.99, 1.00]	0.508	0.85

† Post hoc: defined after the held-out data had been examined.

S6 Claim governance

Sensitivity of the promoted set and of its held-out enrichment to the BFDR thresholds and to the tier rules, and the label-free deconcentration of the promoted set (Sections 3.8 and 5.6).

Table S24: Sensitivity of the promoted set to the governance rules: size, Jaccard index with the primary promoted set, and TF-matched held-out O/E (ENCODE TFBS, 95% TF-cluster bootstrap interval).

	Immune			Kidney			Lung		
Rule	n	Jaccard	O/E	n	Jaccard	O/E	n	Jaccard	O/E
Threshold grid									
$q_M = 0.10$	363	0.74	1.63 [1.27, 2.31]	188	0.68	1.24 [1.04, 1.72]	282	0.68	2.23 [1.76, 3.03]
$q_M = 0.20$	493	1.00	1.67 [1.35, 2.38]	276	1.00	1.24 [1.03, 1.81]	412	1.00	2.27 [1.78, 3.37]
$q_M = 0.30$	596	0.83	1.66 [1.40, 2.35]	350	0.79	1.22 [1.04, 1.81]	538	0.77	2.16 [1.68, 3.02]
$q_M = 0.40$	755	0.65	1.58 [1.36, 2.06]	443	0.62	1.18 [1.02, 1.64]	666	0.62	2.07 [1.71, 2.76]
Dual-universe rule									
primary	493	1.00	1.67 [1.35, 2.38]	276	1.00	1.24 [1.03, 1.81]	412	1.00	2.27 [1.78, 3.37]
lenient (+1 tier)	493	1.00	1.67 [1.35, 2.38]	276	1.00	1.24 [1.03, 1.81]	412	1.00	2.27 [1.78, 3.37]
strict (−1 tier)	341	0.69	1.59 [1.26, 2.26]	186	0.67	1.24 [1.04, 1.69]	248	0.60	2.21 [1.75, 3.19]
Single fit									
global fit only	506	0.97	1.64 [1.35, 2.28]	282	0.98	1.20 [1.01, 1.76]	423	0.97	2.22 [1.77, 3.12]
global fit, Strong only	304	0.62	1.62 [1.28, 2.22]	156	0.57	1.22 [1.03, 1.74]	220	0.53	2.16 [1.74, 2.94]
context fit only	553	0.89	1.65 [1.36, 2.31]	326	0.85	1.24 [1.04, 1.76]	464	0.89	2.31 [1.83, 3.41]
context fit, Strong only	334	0.68	1.59 [1.26, 2.22]	179	0.65	1.27 [1.08, 1.82]	232	0.56	2.27 [1.70, 3.36]
Rank budget									
rank budget	2,008	0.25	1.42 [1.27, 1.77]	–	–	–	–	–	–
Credibility									
NUTS mean (primary)	493	1.00	1.67 [1.35, 2.38]	276	1.00	1.24 [1.03, 1.81]	412	1.00	2.27 [1.78, 3.37]
posterior mode	484	0.98	1.66 [1.36, 2.33]	274	0.97	1.23 [1.04, 1.81]	409	0.99	2.28 [1.79, 3.32]
Estimator									
EM baseline	1,659	0.30	1.45 [1.29, 1.82]	1,051	0.26	1.12 [1.00, 1.33]	1,608	0.26	1.71 [1.52, 2.11]

Table S24: Sensitivity of the promoted set to the governance rules: size, Jaccard index with the primary promoted set, and TF-matched held-out O/E (ENCODE TFBS, 95% TF-cluster bootstrap interval). (continued)

	Immune			Kidney			Lung		
Rule	*n*	Jaccard	O/E	*n*	Jaccard	O/E	*n*	Jaccard	O/E
Simple rules									
majority (≥2 reports)	1,102	0.40	1.62 [1.41, 2.10]	720	0.36	1.16 [1.02, 1.45]	1,079	0.35	1.84 [1.54, 2.41]
union (≥1 report)	2,216	0.22	1.44 [1.28, 1.78]	1,439	0.19	1.09 [0.99, 1.28]	2,178	0.19	1.65 [1.51, 2.08]
DoRothEA A/B	269	0.50	1.67 [1.44, 2.15]	165	0.52	1.16 [0.98, 1.69]	181	0.39	2.04 [1.46, 3.13]
CollecTRI member	2,041	0.24	1.44 [1.26, 1.81]	1,289	0.21	1.11 [1.00, 1.27]	2,022	0.20	1.67 [1.51, 2.01]
Size-matched to primary									
joint, NUTS†	493	1.00	1.68 [1.35, 2.36]	–	–	–	–	–	–
joint, mode†	493	0.96	1.69 [1.36, 2.41]	–	–	–	–	–	–
joint, EM†	493	0.35	1.77 [1.50, 2.44]	–	–	–	–	–	–
global fit, NUTS†	493	0.96	1.63 [1.31, 2.34]	–	–	–	–	–	–
context fit, NUTS†	493	0.92	1.68 [1.39, 2.36]	–	–	–	–	–	–
number of reports†	493	0.48	1.75 [1.48, 2.24]	–	–	–	–	–	–
DoRothEA level†	493	0.48	1.69 [1.46, 2.29]	–	–	–	–	–	–
CollecTRI effort†	493	0.13	1.39 [1.18, 1.84]	–	–	–	–	–	–

Threshold grid: moderate threshold q_M at the primary strong threshold $q_S = 0.05$ (the promoted set does not depend on q_S).
* Few TF clusters.
† Post hoc: defined after the held-out data had been examined.

Table S25: Sensitivity of the promoted set to the governance rules: size, Jaccard index with the primary promoted set, and TF-matched held-out O/E (Pooled perturbation, 95% TF-cluster bootstrap interval).

	Immune			Kidney			Lung		
Rule	n	Jaccard	O/E	n	Jaccard	O/E	n	Jaccard	O/E
Threshold grid									
$q_M = 0.10$	363	0.74	1.23* [0.69, 1.88]	188	0.68	0.98* [0.70, 1.60]	282	0.68	1.26* [1.12, 1.44]
$q_M = 0.20$	493	1.00	1.20 [0.82, 1.64]	276	1.00	0.66* [0.00, 1.00]	412	1.00	1.29* [1.13, 1.57]
$q_M = 0.30$	596	0.83	1.18 [0.86, 1.58]	350	0.79	0.77* [0.00, 0.94]	538	0.77	1.24* [1.08, 1.44]
$q_M = 0.40$	755	0.65	1.21 [0.86, 1.77]	443	0.62	0.85* [0.00, 1.34]	666	0.62	1.20* [0.86, 1.49]
Dual-universe rule									
primary	493	1.00	1.20 [0.82, 1.64]	276	1.00	0.66* [0.00, 1.00]	412	1.00	1.29* [1.13, 1.57]
lenient (+1 tier)	493	1.00	1.20 [0.82, 1.64]	276	1.00	0.66* [0.00, 1.00]	412	1.00	1.29* [1.13, 1.57]
strict (-1 tier)	341	0.69	1.18* [0.66, 1.81]	186	0.67	1.11* [0.84, 1.60]	248	0.60	1.14* [0.95, 1.44]
Single fit									
global fit only	506	0.97	1.16 [0.80, 1.60]	282	0.98	0.66* [0.00, 1.00]	423	0.97	1.29* [1.13, 1.57]
global fit, Strong only	304	0.62	1.17* [0.68, 1.94]	156	0.57	1.51* [1.05, 2.67]	220	0.53	0.99* [0.81, 1.44]
context fit only	553	0.89	1.13 [0.72, 1.55]	326	0.85	0.61* [0.00, 0.80]	464	0.89	1.32* [1.18, 1.57]
context fit, Strong only	334	0.68	1.18* [0.66, 1.81]	179	0.65	1.11* [0.84, 1.60]	232	0.56	1.06* [0.81, 1.44]
Rank budget									
rank budget	2,008	0.25	1.14 [0.96, 1.37]	–	–	–	–	–	–
Credibility									
NUTS mean (primary)	493	1.00	1.20 [0.82, 1.64]	276	1.00	0.66* [0.00, 1.00]	412	1.00	1.29* [1.13, 1.57]
posterior mode	484	0.98	1.21 [0.85, 1.66]	274	0.97	0.66* [0.00, 1.00]	409	0.99	1.29* [1.13, 1.57]
Estimator									
EM baseline	1,659	0.30	1.17 [0.98, 1.41]	1,051	0.26	1.01* [0.81, 2.41]	1,608	0.26	1.23 [0.84, 1.39]

Table S25: Sensitivity of the promoted set to the governance rules: size, Jaccard index with the primary promoted set, and TF-matched held-out O/E (Pooled perturbation, 95% TF-cluster bootstrap interval). (continued)

	Immune			Kidney			Lung		
Rule	*n*	Jaccard	O/E	*n*	Jaccard	O/E	*n*	Jaccard	O/E
Simple rules									
majority (≥2 reports)	1,102	0.40	1.08 [0.75, 1.39]	720	0.36	1.25* [0.61, 4.03]	1,079	0.35	1.17* [0.72, 1.40]
union (≥1 report)	2,216	0.22	1.09 [0.92, 1.30]	1,439	0.19	1.10* [0.68, 3.13]	2,178	0.19	1.16 [0.91, 1.32]
DoRothEA A/B	269	0.50	1.03* [0.48, 1.75]	165	0.52	0.97* [0.84, 1.14]	181	0.39	1.03* [0.85, 1.44]
CollecTRI member	2,041	0.24	1.13 [0.96, 1.32]	1,289	0.21	1.11* [0.70, 2.70]	2,022	0.20	1.19 [0.95, 1.33]
Size-matched to primary									
joint, NUTS†	493	1.00	1.21 [0.85, 1.66]	–	–	–	–	–	–
joint, mode†	493	0.96	1.19 [0.77, 1.62]	–	–	–	–	–	–
joint, EM†	493	0.35	1.16 [0.66, 1.45]	–	–	–	–	–	–
global fit, NUTS†	493	0.96	1.15 [0.75, 1.62]	–	–	–	–	–	–
context fit, NUTS†	493	0.92	1.12 [0.54, 1.55]	–	–	–	–	–	–
number of reports†	493	0.48	1.16 [0.78, 1.42]	–	–	–	–	–	–
DoRothEA level†	493	0.48	0.99 [0.54, 1.45]	–	–	–	–	–	–
CollecTRI effort†	493	0.13	1.35 [1.07, 1.74]	–	–	–	–	–	–

Threshold grid: moderate threshold q_M at the primary strong threshold $q_S = 0.05$ (the promoted set does not depend on q_S).
* Few TF clusters.
† Post hoc: defined after the held-out data had been examined.

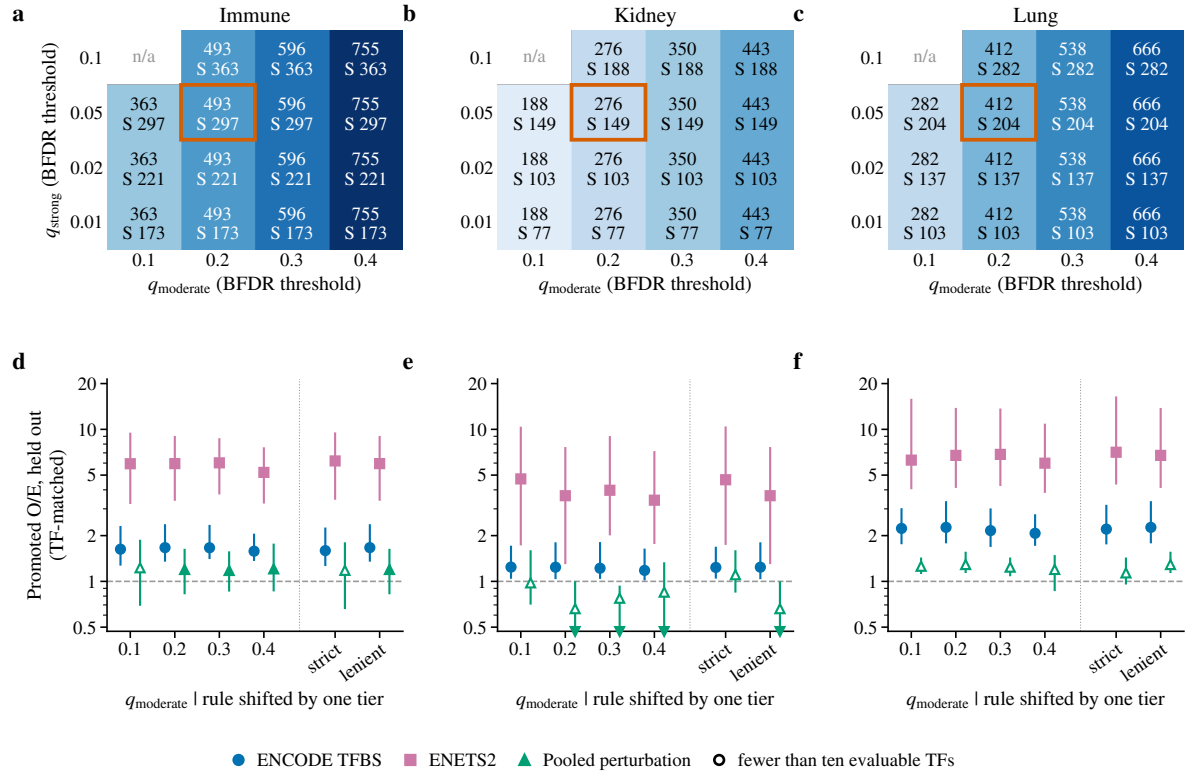


Figure S5: Sensitivity of claim governance to the BFDR thresholds (dual-universe rule). (a–c) Number of promoted edges (Strong ∪ Moderate; colour and upper number) and of Strong edges (S) over the grid of $q_{strong} \times q_{moderate}$; the prespecified point is outlined. Under the primary rule q_{strong} only moves edges between Strong and Moderate, so the promoted set and its enrichment do not depend on it (largest change of O/E across q_{strong}: 0.00). (d–f) Observed over TF-matched expected held-out positives in the promoted set, by $q_{moderate}$ at the prespecified q_{strong}, and for the rules shifted by one tier (strict, lenient) at the prespecified thresholds; bars: 95% TF-cluster bootstrap intervals (triangles: lower limit below the axis range); open markers: intervals resting on fewer than ten evaluable TFs; reference line at 1.

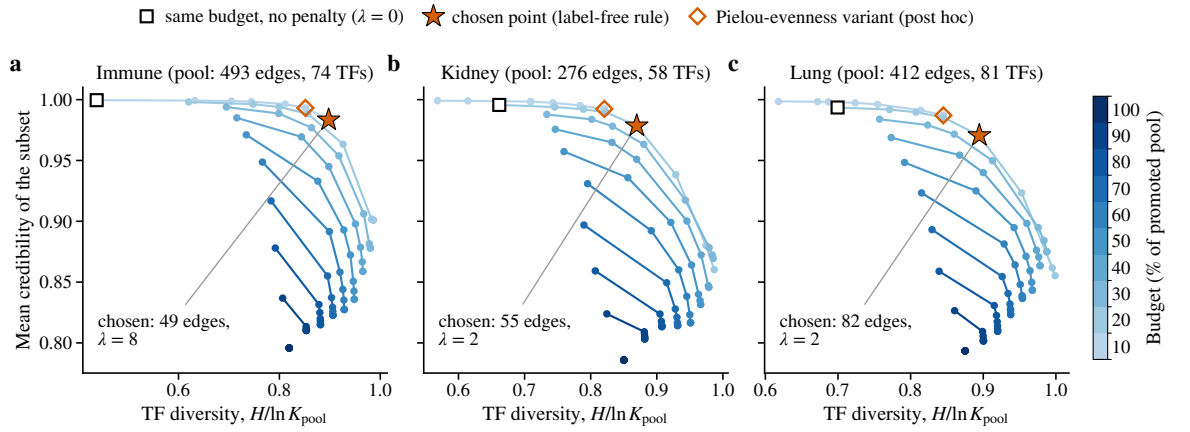


Figure S6: Label-free deconcentration of the promoted set. Each line is one budget (share of the promoted pool, colour) traced over the TF penalty $\lambda \in \{0, 0.25, 0.5, 1, 2, 4, 8, 16\}$; x: TF diversity (Shannon entropy over TFs divided by the log number of TFs in the pool), y: mean credibility of the selected subset (min of global-fit and context-fit credibility). Star: operating point chosen by the label-free rule of Section 3.8 (harmonic mean of normalised credibility and diversity; fixed after held-out results of other analyses existed, but no held-out data enter the choice); open square: the same budget without penalty; open diamond: the post hoc variant with Pielou evenness.

Table S26: Deconcentration of the promoted set: the label-free chosen subset, an undeconcentrated subset of the same size (highest credibility), and the full promoted pool; TF concentration and TF-matched held-out O/E (95% TF-cluster bootstrap interval).

Set	Size	α	Mean $\bar{c}$	TFs	Top TF	Evenness	ENCODE TFBS	ENETS2	Pert.
Immune									
chosen	49	8	0.983	48	0.04	1.00	2.88 [2.05, 4.01]	8.33 [0.00, 26.58]	2.49 [0.00, 3.33]
same size, by credibility	49	0	1.000	12	0.47	0.76	2.16 [1.69, 2.73]	3.16 [0.00, 16.27]	1.85 [1.40, 2.21]
promoted pool	493	0	0.796	74	0.15	0.82	1.67 [1.35, 2.38]	5.95 [3.39, 9.05]	1.20 [0.82, 1.64]
chosen (Pielou criterion)†	49	4	0.993	41	0.04	0.99	2.50 [1.86, 3.47]	2.85 [0.00, 13.10]	2.20 [1.09, 3.32]
same size (Pielou)†	49	0	1.000	12	0.47	0.76	2.16 [1.69, 2.73]	3.16 [0.00, 16.27]	1.85 [1.40, 2.21]
Kidney									
chosen	55	2	0.979	38	0.05	0.97	2.02 [1.41, 2.77]	14.22 [0.00, 31.34]	–
same size, by credibility	55	0	0.996	22	0.24	0.87	1.64 [1.30, 2.47]	6.67 [2.52, 15.14]	1.67 [0.00, 2.11]
promoted pool	276	0	0.786	58	0.14	0.85	1.24 [1.03, 1.81]	3.66 [1.30, 7.65]	0.66 [0.00, 1.00]
chosen (Pielou criterion)†	28	8	0.992	28	0.04	1.00	2.14 [1.15, 3.56]	14.04 [0.00, 49.24]	–
same size (Pielou)†	28	0	0.999	13	0.21	0.90	1.93 [1.37, 2.64]	4.87 [0.00, 16.25]	–
Lung									
chosen	82	2	0.971	56	0.05	0.98	2.72 [2.12, 3.59]	14.24 [3.05, 43.28]	1.23 [1.08, 1.44]
same size, by credibility	82	0	0.994	29	0.13	0.91	2.44 [2.03, 3.49]	6.72 [2.35, 24.71]	0.85 [0.00, 1.44]
promoted pool	412	0	0.793	81	0.08	0.88	2.27 [1.78, 3.37]	6.74 [4.12, 13.83]	1.29 [1.13, 1.57]
chosen (Pielou criterion)†	41	8	0.987	41	0.02	1.00	1.68 [0.46, 2.87]	0.00 [0.00, 0.00]	0.00
same size (Pielou)†	41	0	0.999	19	0.15	0.92	2.61 [2.14, 3.91]	6.61 [0.00, 26.88]	1.46 [0.00, 1.62]

α: TF penalty of the greedy selection (0: no deconcentration). Top TF: share of the set's edges belonging to its most frequent TF; evenness: Pielou's evenness of the TF counts.

† Post hoc: defined after the held-out data had been examined.

S7 Hub-bias simulation

Results of the post hoc hub-bias simulation with known truth described in Section 3.5 (Section 5.7): set sizes, error rates, precision and over-promotion for every estimator and scenario, and the posterior-mode structure of the latent-class fits.

Table S27: Hub-bias simulation with known truth: BFDR ≤ 0.05 set size and realised FDR, precision and over-promotion index of top-popularity-decile TFs at a set size matched to the Family-HB set, and AUROC over all edges (mean over replicates, 95% t interval).[†]

	BFDR ≤ 0.05 set		Size-matched set		
Method	Size	FDR	Precision [95% CI]	Index [95% CI]	AUROC
Truth independent of popularity					
Oracle posterior	2,902	0.07	0.98 [0.97, 0.99]	4.65 [4.31, 5.00]	0.909
Family-HB	1,094	0.07	0.93 [0.92, 0.95]	5.09 [4.57, 5.61]	0.858
EM baseline	5,518	0.23	0.89 [0.87, 0.90]	6.62 [5.94, 7.31]	0.858
Dependence-aware	201	0.10	0.79 [0.74, 0.84]	3.38 [2.70, 4.07]	0.856
Family-regularized	5,517	0.23	0.89 [0.87, 0.90]	6.62 [5.94, 7.31]	0.858
Union	–	–	0.38 [0.37, 0.39]	1.98 [1.88, 2.08]	0.803
Majority ($\geq$2)	–	–	0.72 [0.71, 0.73]	2.58 [2.49, 2.68]	0.682
Vote fraction	–	–	0.61 [0.58, 0.64]	1.44 [1.12, 1.77]	0.805
Fisher	–	–	0.90 [0.88, 0.91]	6.29 [5.62, 6.95]	0.839
Stouffer	–	–	0.89 [0.87, 0.91]	5.99 [5.35, 6.62]	0.836
Family-HB + observation (n = 5)	115,475	1.00	0.00 [0.00, 0.00]	7.95 [5.43, 10.46]	0.255
EM + observation	55,172	0.70	0.86 [0.84, 0.89]	6.43 [5.77, 7.10]	0.817
Biological hubs					
Oracle posterior	4,671	0.07	0.98 [0.97, 0.99]	3.36 [3.10, 3.63]	0.923
Family-HB	1,842	0.08	0.92 [0.89, 0.94]	2.99 [2.62, 3.36]	0.886
EM baseline	7,635	0.24	0.92 [0.91, 0.94]	3.95 [3.57, 4.33]	0.887
Dependence-aware	168	0.02	0.90 [0.86, 0.93]	3.02 [2.59, 3.45]	0.884
Family-regularized	7,634	0.24	0.92 [0.91, 0.94]	3.95 [3.57, 4.33]	0.887
Union	–	–	0.42 [0.41, 0.43]	1.34 [1.28, 1.39]	0.849
Majority ($\geq$2)	–	–	0.77 [0.76, 0.78]	1.89 [1.82, 1.96]	0.737
Vote fraction	–	–	0.70 [0.65, 0.74]	1.59 [1.38, 1.80]	0.851
Fisher	–	–	0.92 [0.91, 0.94]	3.80 [3.42, 4.18]	0.876
Stouffer	–	–	0.92 [0.90, 0.94]	3.69 [3.33, 4.04]	0.873
Family-HB + observation (n = 4)	146,534	1.00	0.00 [0.00, 0.00]	3.04 [2.02, 4.06]	0.208
EM + observation	85,709	0.87	0.90 [0.88, 0.92]	3.88 [3.52, 4.24]	0.835
Hub-concentrated literature copying					
Oracle posterior	1,195	0.06	0.96 [0.95, 0.97]	2.69 [2.55, 2.84]	0.907
Family-HB	784	0.23	0.77 [0.70, 0.84]	5.34 [4.70, 5.97]	0.864
EM baseline	10,778	0.54	0.41 [0.36, 0.45]	9.01 [8.63, 9.38]	0.863
Dependence-aware	172	0.47	0.60 [0.55, 0.66]	5.44 [4.34, 6.53]	0.862
Family-regularized	10,775	0.54	0.41 [0.36, 0.45]	9.01 [8.63, 9.38]	0.863
Union	–	–	0.31 [0.30, 0.32]	2.61 [2.47, 2.76]	0.799
Majority ($\geq$2)	–	–	0.46 [0.45, 0.47]	3.94 [3.76, 4.12]	0.679
Vote fraction	–	–	0.47 [0.46, 0.48]	1.33 [0.96, 1.71]	0.800
Fisher	–	–	0.46 [0.42, 0.50]	8.59 [8.20, 8.98]	0.848
Stouffer	–	–	0.48 [0.43, 0.52]	8.20 [7.78, 8.62]	0.846
Family-HB + observation (n = 3)	79,154	1.00	0.00 [0.00, 0.00]	6.17 [3.28, 9.07]	0.267
EM + observation	11,770	0.55	0.42 [0.37, 0.46]	9.00 [8.63, 9.38]	0.873

Over-promotion index: share of the set at top-popularity-decile TFs divided by the share of true edges there (1 = proportional to biology). n: replicates behind a row when fewer than the design. Precision intervals are truncated at 0.

[†] Post hoc: defined after the held-out data had been examined.

Table S28: Posterior-mode structure of the latent-class fits in the hub-bias simulation: restarts reaching the best optimum, distinct optima, median gap to the next optimum, selected optima that are label-swapped or have an inflated true class, and the fitted class weight.[†]

Model	Reps	Restarts	At best	Optima	Gap	Swapped	Inflated	w_T	Truth
Truth independent of popularity									
EM baseline	20	10	1.00	1.0	–	0	0	0.01	0.0099
Dependence-aware	20	10	0.87	1.4	3.3	13	0	0.0017	0.0099
Family-regularized	20	10	0.98	1.1	14814.4	0	0	0.01	0.0099
Family-HB	20	30	0.43	4.1	10.8	0	0	0.0085	0.0099
EM + observation	20	10	0.43	1.8	8129.8	0	14	0.38	0.0099
Family-HB + observation	5	20	0.49	4.8	4310.6	5	5	0.50	0.01
Biological hubs									
EM baseline	20	10	1.00	1.0	–	0	0	0.01	0.01
Dependence-aware	20	10	0.99	1.1	6.4	5	0	0.0043	0.01
Family-regularized	20	10	0.94	1.6	27797.1	0	0	0.01	0.01
Family-HB	20	30	0.48	3.4	12.4	0	0	0.0098	0.01
EM + observation	20	10	0.27	1.9	11950.6	0	19	0.36	0.01
Family-HB + observation	4	20	0.64	3.5	3464.2	4	4	0.42	0.0099
Hub-concentrated literature copying									
EM baseline	20	10	1.00	1.0	–	0	0	0.02	0.0098
Dependence-aware	20	10	0.93	1.2	14.3	1	0	0.0038	0.0098
Family-regularized	20	10	0.96	1.4	33702.0	0	0	0.02	0.0098
Family-HB	20	30	0.56	3.4	60.8	0	0	0.0087	0.0098
EM + observation	20	10	0.90	1.6	4692.2	0	1	0.01	0.0098
Family-HB + observation	3	20	0.52	3.0	6881.6	3	3	0.37	0.0097

[†] Post hoc: defined after the held-out data had been examined.

S8 Replication check

The pooled Bayesian replication model of Section 3.9: per-study posterior summaries, dependence on the priors, and the leave-k-out refits (Section 5.8).

Table S29: Pooled replication model under the primary prior (study level): per-study odds ratio $\exp(\theta_u)$ (posterior median, 95% CrI) and pass probability, beside the per-study frequentist estimates.

Study	OR [95% CrI]	P(pass)	O/E	MH OR	Observed	Expected
Dixit 2016	2.74 [0.84, 5.23]	0.89	0.00	0.00	0	0.21
Norman 2019	3.07 [2.37, 3.96]	1.00	1.95	3.27	100	51
Papalexi 2021	3.46 [2.64, 4.69]	1.00	2.24	3.64	84	38
Replogle 2022	2.43 [1.54, 3.42]	0.98	1.57	2.07	38	24
Pooled, $\exp(\mu)$	2.82 [1.42, 4.09]	0.97	–	–	–	–
New study, $\exp(\theta_{\text{new}})$	2.89 [0.92, 6.56]	0.92	–	–	–	–

Between-study SD σ: 0.32 [0.02, 0.88]; max. $\widehat{R}$ 1.007, min. bulk ESS 1,286, 0 divergent transitions.

Table S30: Prior sensitivity of the pooled replication model (study level): pooled odds ratio $\exp(\mu)$ and odds ratio of a new study $\exp(\theta_{\text{new}})$ (posterior median, 95% CrI), between-study SD σ, posterior probability that the pooled odds ratio reaches the endpoint, and each study's posterior probability of passing.

Profile	$\exp(\mu)$	P(end-	σ	$\exp(\theta_{\text{new}})$	Pass probability			
(μ, σ priors)		point)			Dixit	Norman	Papalexi	Replogle
primary	2.82	0.97	0.32	2.89	0.89	1.00	1.00	0.98
N(0, 1), HN(0.5)	[1.42, 4.09]		[0.02, 0.88]	[0.92, 6.56]				
mu sd0.5	2.43	0.90	0.41	2.53	0.79	1.00	1.00	0.97
N(0, 0.5), HN(0.5)	[1.11, 3.50]		[0.03, 1.04]	[0.57, 6.12]				
mu sd2	2.90	0.98	0.31	2.92	0.91	1.00	1.00	0.98
N(0, 2), HN(0.5)	[1.55, 4.36]		[0.02, 0.85]	[1.06, 6.85]				
sigma hn0.25	2.91	0.99	0.21	2.92	0.96	1.00	1.00	0.99
N(0, 1), HN(0.25)	[1.97, 3.92]		[0.01, 0.51]	[1.43, 5.06]				
sigma hn1	2.72	0.92	0.47	2.78	0.81	1.00	1.00	0.97
N(0, 1), HN(1)	[1.01, 4.27]		[0.04, 1.48]	[0.49, 8.78]				
skeptical	1.21	0.15	0.77	1.28	0.33	1.00	1.00	0.91
N(0, 0.2), HN(0.5)	[0.81, 1.81]		[0.38, 1.28]	[0.21, 6.42]				
skeptical sigma0.25	1.38	0.34	0.52	1.43	0.41	1.00	1.00	0.90
N(0, 0.2), HN(0.25)	[0.92, 2.06]		[0.26, 0.82]	[0.42, 4.04]				
diffuse	2.87	0.96	0.44	2.88	0.84	1.00	1.00	0.97
N(0, 2), HN(1)	[1.28, 4.79]		[0.04, 1.35]	[0.55, 9.07]				
tf random slope†	3.28	0.99	0.29	3.31	0.93	1.00	1.00	1.00
N(0, 1), HN(0.5)	[1.71, 5.42]		[0.01, 0.91]	[1.02, 7.99]				

N, HN: normal and half-normal priors of μ and σ. Pass probability: posterior probability that the study's odds ratio reaches the endpoint. tf random slope: adds a TF-level random slope. Max. $\widehat{R}$ 1.007 under the primary prior.
† Post hoc: defined after the held-out data had been examined.

Table S31: Leave-k-out refits of the pooled replication model: changes in the pooled log odds ratio μ, in the posterior probability that the pooled odds ratio reaches the endpoint, and in the per-unit pass probabilities, over all subsets of k units left out.

		$\lvert\Delta\mu\rvert$		$\lvert\Delta P\rvert$		$\lvert\Delta$ pass prob.$\rvert$	
Scheme	Fits	mean	max	mean	max	mean	max
Study level (full data: μ = 1.00, P = 0.97)							
L1O	4	0.10	0.15	0.03	0.06	0.02	0.09
L2O	6	0.16	0.45	0.10	0.29	0.05	0.26
L3O	4	0.45	1.24	0.27	0.72	0.17	0.63
Cohort level (full data: μ = 1.01, P = 0.97)							
L1O	5	0.09	0.17	0.03	0.07	0.02	0.11
L2O	10	0.14	0.46	0.07	0.28	0.05	0.26
L3O	10	0.28	1.25	0.17	0.74	0.12	0.66

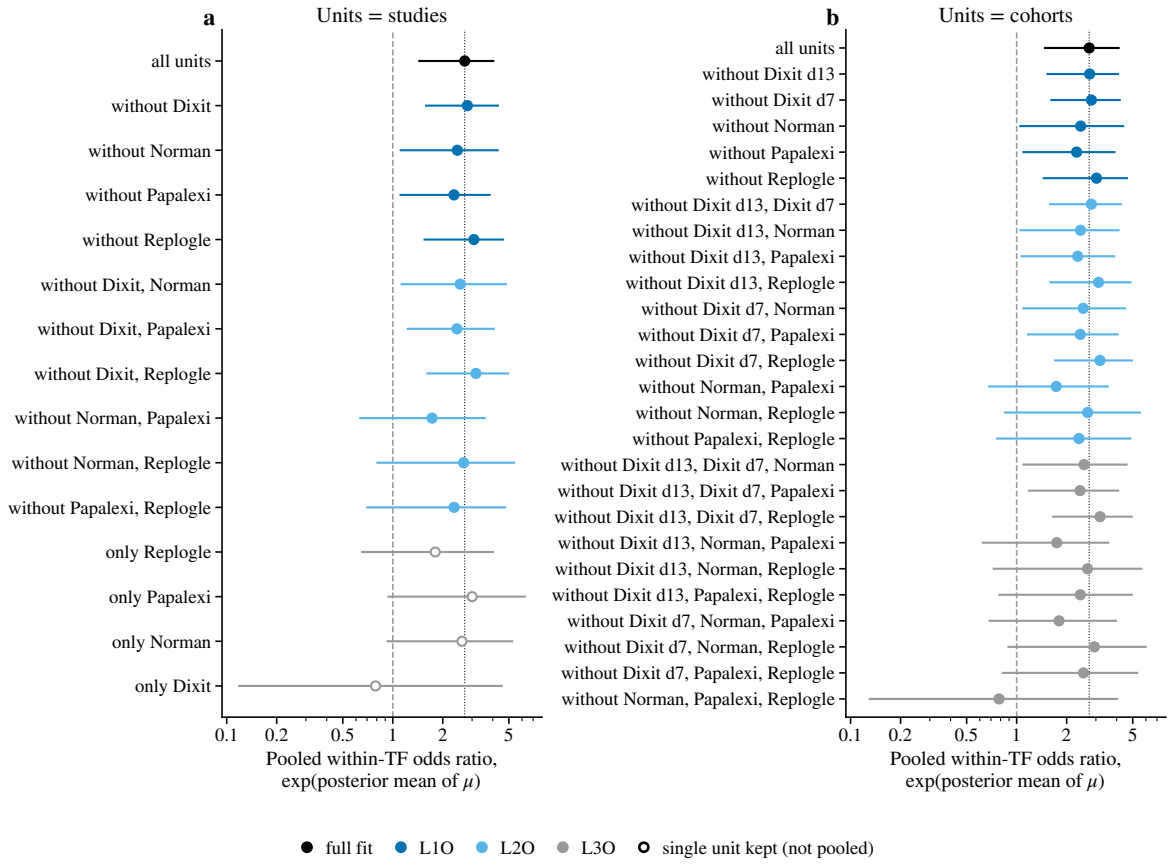


Figure S7: Leave-k-out drift of the pooled replication effect. Points: exp of the posterior mean of the pooled within-TF log odds ratio μ of the Bayesian random-effects model, with the exponentiated 95% credible limits (prior $\mu \sim N(0, 1^2)$, $\sigma \sim HN(0.5)$), for the full data and refitted after removing every subset of one (L1O), two (L2O) or three (L3O) units, with studies (a) or cohorts (b) as units; the main text reports the posterior median of $\exp(\mu)$, which is 2.82 at study level against 2.71 for the mean shown here. Open markers: fits that keep a single unit, whose μ is that unit's effect shrunk towards the prior rather than a pooled estimate. Dashed line: no enrichment; dotted line: full-data posterior mean.

S9 Computation

Scaling exponents measured on synthetic evidence and the complexity of each computational stage (Section 5.10).

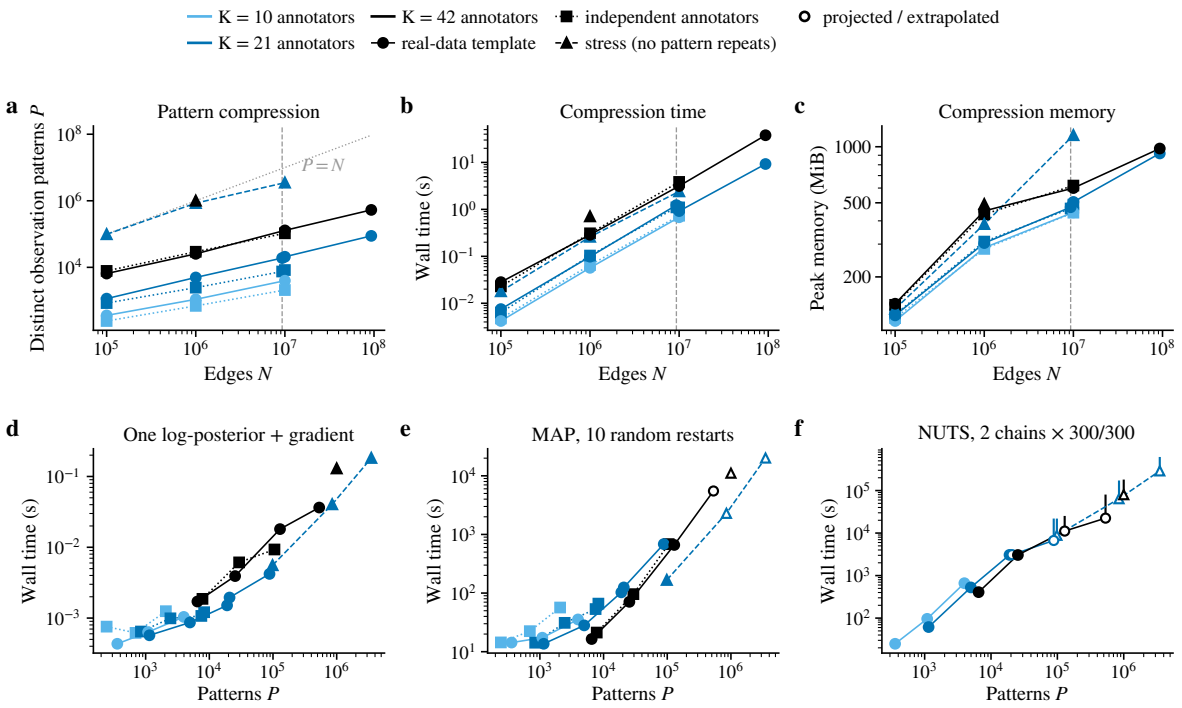


Figure S8: Computational scaling on synthetic evidence (no held-out data). (a) Number of distinct observation patterns P against the number of edges N for evidence generated from the real observation design (template), from independent annotators and without repeated patterns (stress); dashed vertical line: size of the global universe. (b, c) Wall time and peak memory of the streaming pattern compression. (d) One evaluation of the collapsed log posterior and its gradient. (e) MAP with 10 random restarts (open: extrapolated from the restarts that finished, or projected from evaluation counts, a lower bound). (f) NUTS with 2 chains of 300 warm-up and 300 draws from random starts, a cost benchmark rather than a converged fit; open markers are projections from the steps per iteration of the largest measured run (lowest over the two measurement sessions), which are lower bounds because the steps per iteration grow with N, and the vertical bars reach the projection at the tree-depth cap (highest over the sessions), the better estimate at ten times the real size; the same ranges as in Table 15. Measured times are wall-clock on a shared laptop (Apple M2 Pro, 10 cores, 32 GiB) and are upper bounds of the undisturbed cost.

Table S32: Empirical scaling exponents from the compute-scaling experiment: least-squares fits of $\log_{10} y = a + b \log_{10} x$ over the synthetic configurations (slope b with standard error, R^2, number of configurations).

Quantity y vs x	Configurations	n	Slope (SE)	R^2
Measurement session a				
P vs N	independent streaming compression $K = 10$	3	0.47 (0.01)	1.00
P vs N	independent streaming compression $K = 21$	4	0.49 (0.01)	1.00
P vs N	independent streaming compression $K = 42$	3	0.57 (0.01)	1.00
P vs N	template streaming compression $K = 10$	3	0.52 (0.02)	1.00
P vs N	template streaming compression $K = 21$	5	0.63 (0.01)	1.00
P vs N	template streaming compression $K = 42$	4	0.65 (0.01)	1.00
P vs N	template whole-matrix compression $K = 21$	4	0.62 (0.01)	1.00
compression wall time vs N	streaming compression, all K and generators	22	1.09 (0.07)	0.93
compression wall time vs NK	streaming compression, all K and generators	22	1.07 (0.02)	0.99
peak RSS vs N	streaming compression, all K and generators	22	0.29 (0.02)	0.92
compression wall time vs N	whole-matrix compression, all K and generators	4	1.19 (0.06)	0.99
compression wall time vs NK	whole-matrix compression, all K and generators	4	1.19 (0.06)	0.99
peak RSS vs N	whole-matrix compression, all K and generators	4	0.45 (0.03)	0.99
log posterior and gradient time vs PKS	all	26	0.56 (0.04)	0.90
L-BFGS evaluation time vs PKS	all	25	0.39 (0.03)	0.87
MAP, 10 restarts, time vs PKS	all, measured or extrapolated	25	0.56 (0.05)	0.84
MAP peak RSS vs PK	all	26	0.25 (0.03)	0.75
log posterior and gradient time vs PKS	$P \geq 10^4$	12	0.84 (0.06)	0.95
L-BFGS evaluation time vs PKS	$P \geq 10^4$	11	0.63 (0.05)	0.95
MAP, 10 restarts, time vs PKS	$P \geq 10^4$, measured or extrapolated	11	0.90 (0.12)	0.87
MAP peak RSS vs PK	$P \geq 10^4$	12	0.50 (0.04)	0.93
MAP evaluations per restart vs N	template and independent generators, restarts run	22	0.42 (0.02)	0.95
MAP evaluations per restart vs P	template and independent generators, restarts run	22	0.40 (0.06)	0.71
NUTS time per leapfrog step (compile excluded) vs PKS	measured	8	0.80 (0.05)	0.98
NUTS, 2 chains $\times$ (300 + 300) wall time vs PKS	measured	8	0.81 (0.14)	0.85
NUTS leapfrog steps per iteration vs N	measured	8	0.36 (0.03)	0.96

Table S32: Empirical scaling exponents from the compute-scaling experiment: least-squares fits of $\log_{10} y = a + b \log_{10} x$ over the synthetic configurations (slope b with standard error, R^2, number of configurations). (continued)

Quantity y vs x	Configurations	n	Slope (SE)	R^2
Measurement session b[†]				
compression wall time vs N	streaming compression, all K and generators	22	0.90 (0.10)	0.82
compression wall time vs NK	streaming compression, all K and generators	22	0.88 (0.08)	0.87
peak RSS vs N	streaming compression, all K and generators	22	0.28 (0.02)	0.93
compression wall time vs N	whole-matrix compression, all K and generators	4	0.86 (0.08)	0.98
compression wall time vs NK	whole-matrix compression, all K and generators	4	0.86 (0.08)	0.98
peak RSS vs N	whole-matrix compression, all K and generators	4	0.44 (0.06)	0.97
log posterior and gradient time vs PKS	all	26	0.51 (0.05)	0.81
log posterior and gradient time (minimum) vs PKS	all	26	0.61 (0.04)	0.90
L-BFGS evaluation time vs PKS	all	19	0.23 (0.04)	0.71
MAP, 10 restarts, time vs PKS	all, measured or extrapolated	19	0.29 (0.07)	0.54
MAP peak RSS vs PK	all	26	0.23 (0.02)	0.79
log posterior and gradient time vs PKS	$P \geq 10^4$	12	0.92 (0.06)	0.96
log posterior and gradient time (minimum) vs PKS	$P \geq 10^4$	12	0.93 (0.06)	0.96
L-BFGS evaluation time vs PKS	$P \geq 10^4$	5	0.39 (0.06)	0.93
MAP, 10 restarts, time vs PKS	$P \geq 10^4$, measured or extrapolated	5	0.03 (0.24)	0.00
MAP peak RSS vs PK	$P \geq 10^4$	12	0.43 (0.04)	0.91
MAP evaluations per restart vs N	template and independent generators, restarts run	18	0.38 (0.02)	0.94
MAP evaluations per restart vs P	template and independent generators, restarts run	18	0.35 (0.09)	0.46
NUTS time per leapfrog step (compile excluded) vs PKS	measured	6	0.58 (0.04)	0.98
NUTS, 2 chains $\times$ (300 + 300) wall time vs PKS	measured	6	0.62 (0.22)	0.67
NUTS leapfrog steps per iteration vs N	measured	6	0.37 (0.04)	0.95

Session b replicated session a on the same machine with one measurement process at a time. Streaming compression: patterns found TF chunk by TF chunk and merged; whole-matrix compression: one pass over the full evidence matrix.
[†] Post hoc: defined after the held-out data had been examined.

Table S33: Time and memory complexity of each stage (N edges, K annotators, P observation patterns, S latent classes; further symbols in the notes column).

Stage	Time	Memory	Notes
Evidence encoding	$O(NK)$	$O(n_c K)$ per chunk	the K values of an edge (unobserved, unreported, reported) are packed into base-3 words
Pattern compression, streaming by TF chunk	$O(NK + N\log n_c + CP_c\log(CP_c))$	$O(N)$ edge-to-pattern index + $O(n_c K)$ chunk + $O(PK)$ table	sort-based unique per chunk of n_c edges, merge of the C chunk dictionaries, $O(N)$ re-indexing; hashing would give $O(NK)$ expected time
Pattern compression, whole matrix	$O(NK\log N)$	$O(NK)$ matrix + $O(N)$ sort arrays	row-wise unique of the full evidence matrix
Number of patterns P	$P \leq \min(N, 3^K, M + N_{\text{rep}})$	–	M distinct observation masks, N_{rep} edges with at least one report: every unreported edge has the all-zero pattern of its mask; TF-blocked observation keeps M small
Log posterior and gradient, one evaluation	$O(PKS)$	$O(PK) + O(PS)$	independent of N given P: N enters only through the pattern counts
MAP, multi-start L-BFGS	$O(RIPKS)$	$O(PK) + O(RP)$	R restarts, I evaluations per restart; I grew with N in the experiments (Table S32)
NUTS	$O(CDLPKS)$	$O(PK + CDd)$, + $O(CDP)$ if credibility is stored per draw	C chains, D iterations, L leapfrog steps per iteration (at most $2^{10} - 1$), d parameters; L grew with N up to the tree-depth cap
Posterior credibility summaries	$O(D'PKS)$	$O(D'P)$ (quantiles); $O(P)$ (streaming mean and SD)	D' retained draws
Edge-level outputs	$O(N)$	$O(N)$	the credibility of an edge is that of its pattern (index lookup)

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